Design and Fabrication of Blood Pressure Monitoring Device Using Pulse Transit Time (PTT) For Commercial Use

Submitted by

Rawas Saad (200012135)

Sadman Sakib Khan (200012126)

Zahin Tasnim Ridita (200012153)

Sajjad Al Sajal (200012142)



CAPSTONE DESIGN II (IPE 4710) [Group 05]

DEPARTMENT OF MECHANICAL AND PRODUCTION ENGINEERING

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

March 2025

EXECUTIVE SUMMARY

The capstone project developed a non-invasive blood pressure (BP) measurement device using Pulse Transit Time (PTT) to address the global hypertension challenge affecting millions, particularly in low-resource settings, where traditional cuff-based methods, though accurate, were impractical for continuous use due to discomfort and logistical issues, and existing cuffless PTT solutions faced calibration dependency, motion artifacts, and drift, limiting accessibility. The objectives were to design a prototype to achieve BP accuracy within a target range, display real-time data, and maintain cost-effectiveness, tackling these challenges through a literature review confirming PTT's inverse BP correlation, system design, simulations for signal validation, and prototype assembly with noise mitigation techniques, followed by testing on a small group against a standard device. The prototype measured PTT and estimated BP with a mean error within the target range, displayed heart rates and PTT in real time, reduced noise by a notable percentage, and stayed within the allocated budget using local resources and university facilities, demonstrating the application of engineering principles (CO1), problem analysis with research-based solutions (CO2), and effective communication (CO10). The project concluded with a viable, affordable BP monitoring solution, offering potential for early hypertension detection in underserved areas, with future work focusing on wearable integration and dynamic calibration to enhance healthcare sustainability and impact.

ACKNOWLEDGEMENTS

Dr. Md. Anayet Ullah Patwari

Professor

Department of Mechanical and Production Engineering (MPE)

Islamic University of Technology

&

Dr. Mohammad Nasim

Assistant Professor

Department of Mechanical and Production Engineering (MPE)

Islamic University of Technology

MAPPING OF ASSESSMENT, COs TO POs

Table 1: Mapping of Course Outcomes (COs) to Program Outcomes (POs)

	CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
ASSESSMENT 1: 1: Engineering Knowledge Application	**CO1**	1.3, 3.1											
ASSESSMENT 2: Problem Identification and Analysis	**CO2**		1.2, 2.1										
ASSESSMENT 3: Design and Development of Solutions	**CO3**			3.2, 6.1									
ASSESSMENT 4: Research and Investigation	**CO4**				2.1, 7.4 11.1, 11.2								
ASSESSMENT 5: Modern Tools and Techniques Application	**CO5**					5.1, 6.3							
ASSESSMENT 6: Safety and Contextual Reasoning	**CO6**						9.1						
ASSESSMENT 7: Sustainability and Environmental Impact	**CO7**							8.2					
ASSESSMENT 8: Ethics and Professional Responsibility	**CO8**								9.2				
ASSESSMENT 9: Teamwork and Leadership	**CO9**									10.2			
ASSESSMENT 10: Effective Communication	**CO10**										1.5 12.1		

ASSESSMENT 11: Project Management and Financial Planning	**CO11**											10.4,	
ASSESSMENT 12: Lifelong Learning and Adaptability	**CO12**												2.3, 12.2, 12.3 11.3

Table of Contents

EXECUTIVE SUMMARY **1**
ACKNOWLEDGEMENTS **2**
MAPPING OF ASSESSMENT, COs TO POs **3**
List of Figures **8**
List of Tables **9**
CHAPTER 1: INTRODUCTION **1**
1.1. Background and Motivation 1
1.2. Problem Statement 1
1.3. Problem Objective 1
1.4. Scope of the Project 2
1.5. Report Organization 2
CHAPTER 2: LITERATURE REVIEW **3**
2.1. Review of Existing Solutions 3
Current Approaches 3
2.2. Relevant Technologies and Methods 4
2.3. Gaps in Current Knowledge 5
2.4. Summary of Findings 6
CHAPTER 3: SYSTEM DESIGN AND METHODOLOGY **6**
3.1 Design Requirements and Specifications 6
3.2 Conceptual Design 7
3.3 Final Design Selection 10
3.4 Design Analysis and Calculations 12
3.5 Tools and Techniques Used 14
CHAPTER 4: COMPONENT SELECTION AND MATERIALS **16**
4.1 Criteria for Component Selection 17
4.2 Materials Used 17
Electronic Parts: 17
Materials Used for Fabrication 18
4.3 Cost Analysis and Budget 19
4.4 Sourcing of Components 20
CHAPTER 5: MODELING AND SIMULATION **20**
5.1 Simulation Software and Tools 20
5.2 Model Development 21
5.3 Simulation Results and Analysis 22
5.4 Design Optimization Based on Simulation 23
CHAPTER 6: PROTOTYPE DEVELOPMENT **24**
6.1 Assembly Process 24
6.2 Challenges Faced During Fabrication 27
6.3 Testing Procedures and Setup 30

CHAPTER 7: EXPERIMENTAL TESTING AND RESULTS 33
7.1 Testing Objectives 33
7.2 Test Plan and Protocol 33
7.3 Data Collection 34
7.4 Analysis of Results 34
7.5 Comparison with Design Specifications 36
CHAPTER 8: EVALUATION OF SUSTAINABILITY AND ENVIRONMENTAL IMPACT 37
8.1 Energy Efficiency Analysis 37
8.2 Environmental Impact Assessment 39
8.3 Sustainable Practices and Recommendations 41
CHAPTER 9: SAFETY, ETHICS, AND PROFESSIONAL RESPONSIBILITY 42
9.1 Safety Considerations in Design and Operation 43
9.2 Ethical Considerations in Project Execution 43
9.3 Compliance with Industry Standards and Regulations 44
CHAPTER 10: PROJECT MANAGEMENT AND TEAM COLLABORATION 45
10.1 Project Planning and Timeline 45
10.2 Team Communication and Coordination 49
10.3 Budget and Resource Management 49
CHAPTER 11: DISCUSSION 51
11.1 Interpretation of Results 51
11.2 Lessons Learned 53
Practical System Integration 53
Technical Skills in Biomedical Signal Processing 53
Adapting to New Technologies 54
11.3 Limitations of the Project 54
Sensor and Signal Limitations 54
Design and Practical Constraints 54
Periodic Calibration Requirement 55
11.4 Potential Improvements 55
Advanced Signal Processing and Calibration 55
Hardware and Sensor Improvements 55
Wearable Integration 55
CHAPTER 12: CONCLUSIONS AND RECOMMENDATIONS 56
12.1 Summary of Key Findings 56
12.2 Recommendations of Future Work 57
12.3 Impact of the Project Outcomes 57
REFERENCES 59
APPENDICES 61
A. Detailed calculations 62
B. Additional figures and tables 63

C. User Manuals and Operational Guidelines 65
D. Raw Data from Experiments 66
E. Relevant Standards and Codes 66
F. Course Outcomes 67

List of Figures

Figure 1: Design of the Wearable Chest ECG Belt 8

Figure 2: CAD Design of the Central Unit Part 9

Figure 3: Final Prototype 11

Figure 4: Project workflow methodology 14

Figure 5: Project Gantt Chart 45

Figure 6: Resource Allocation Pie Chart 50

Figure 7: Pulse time transfer data reading 52

Figure 8: Circuit Diagram of Prototype 64

List of Tables

Table 4.1: Significant Criteria for Materials 15

Table 4.2.1: Electronic Materials For Device 16

Table 4.2.2: Fabrication Materials for Device 17

Table 4.3: Date & List of Purchased Equipments 17

Table 4.4: Source of Purchase 18

Table 7.5: Comparison of Desired and Achieved Parameters 35

Table 8.1: Estimated Power Consumption of Device Components 36

Table 10.4: Bill of Materials 47

Table 11.1: Realtime Data of multiple individuals 50

Table B.1: Component Specifications 61

Table B.2: Test Results Comparison 62

Table D.1: Raw Data from Five Users 63

CHAPTER 1: INTRODUCTION

This project focuses on designing and fabricating a blood pressure measurement device using Pulse Transit Time (PTT), incorporating electrocardiogram (ECG) and photoplethysmography (PPG) methods

1.1. Background and Motivation

The worldwide prevalence of hypertension which affects more than 1.28 billion people demonstrates the necessity for effective blood pressure monitoring especially in Bangladesh and other low- and middle-income countries because they have restricted healthcare access. Cuff-based sphygmomanometers deliver precise readings, yet they cannot provide continuous blood pressure tracking because they cause patient discomfort and present operational difficulties. The cuffless PTT method which measures ECG R-wave to PPG pulse wave time delay has emerged as a promising alternative because it receives support from wearable technology and machine learning advancements. The current solutions encounter multiple obstacles including calibration requirements and motion interference and population-based measurement inaccuracies which prevent their broad market adoption. The movement toward individualized healthcare combined with wearable technology shows promise for developing budget-friendly innovative solutions. The project aims to fill existing gaps by using accessible sensors such as AD8232 and MAX30100 to create a portable BP monitor for underserved communities while promoting health equity and advancing biomedical engineering research at IUT.

1.2. Problem Statement

The current blood pressure monitoring techniques involve invasive procedures and large equipment that cannot be used for continuous non-invasive measurements. The current methods create difficulties for patients who require ongoing monitoring such as those with sleep disorders or hypertension. There's a need for a wearable device that provides accurate, continuous BP estimates without discomfort and includes heart rhythm monitoring for a holistic view, addressing gaps in existing solutions**.**

1.3. Problem Objective

The primary objectives of the project are:

- To design and fabricate a prototype BP monitoring device that measures PTT using ECG (via AD8232) and PPG (via MAX30100) sensors, achieving an accuracy of ±5 mmHg compared to a digital sphygmomanometer.
- To implement a real-time data display system using a 20x4 I2C LCD, showing heart rate and PTT, as validated in testing.
- To develop a cost-effective solution.
- To evaluate and mitigate motion artifacts and signal noise through hardware and software enhancements.
- To explore the feasibility of a wearable design, identifying future improvements for ambulatory monitoring.

1.4. Scope of the Project

This project includes designing a single wearable device with integrated ECG and PPG sensors, developing software for signal processing and real-time display, and conducting initial tests to validate BP estimation accuracy against manual measurements. While initially considering 3D printing, the device was fabricated using conventional methods, and no 3D printed models were made. Exclusions are extensive clinical trials, power efficiency optimization, external system integration, and full commercial development, focusing on a functional prototype.

1.5. Report Organization

This report is structured to provide a comprehensive overview of the blood pressure measurement device project. Chapter 1 introduces the problem, objectives, and scope. Chapter 2 reviews relevant literature and existing technologies. Chapter 3 details the system design and methodology, including design and development processes. Chapter 4 presents the component selection and materials. Chapter 5 discusses modeling and simulation results. Chapter 6 outlines prototype development. Chapter 7 provides experimental testing and results of the prototype testing. Chapter 8 evaluates sustainability and environmental impact. Chapter 9 covers safety, ethics, and professional responsibility. Chapter 10 includes project management and team collaboration. Chapter 11 discusses findings, lessons, limitations, and improvements. Chapter 12 summarizes conclusions and recommendations for future work. References and appendices provide additional technical details and sources, ensuring readers can navigate the document efficiently.

CHAPTER 2: LITERATURE REVIEW

2.1. Review of Existing Solutions

Traditional blood pressure (BP) monitoring relies on cuff-based sphygmomanometers, which, although accurate, are unsuitable for continuous, real-time tracking. To overcome this, cuffless methods based on Pulse Transit Time (PTT) have been explored. PTT is the time delay between the electrical activation of the heart (ECG R-wave) and the arrival of the pulse wave (PPG foot point) at a peripheral site.

Current Approaches

PTT-Based BP Estimation

Block et al. (2016) validated PTT as a marker for BP changes, with an inverse correlation with a mean error of ~5 mmHg post-calibration [2]. Choi et al. (2017) proposed a real-time system using ECG electrodes and a microphone, with errors of 4–6 mmHg, with the strength of cost-effectiveness but the weakness of noise interference [3]. These studies show that PTT can be used for continuous monitoring, but they are limited by calibration dependency and environmental noise.

Wearable Technology for BP Monitoring

SeismoMatch represents a wearable device created by Carek et al. (2017) which combines seismocardiography with PPG technology to measure blood pressure with a 5.2 mmHg mean error but experiences motion artifacts during use [5]. The research by Ganti et al. (2021) concentrated on underserved areas while achieving errors between 4–7 mmHg but they noted calibration drift as a limitation [9]. The system demonstrates portability and user-friendly design but suffers from motion-related inaccuracies.

Alternative Sensing Methods

The contactless UWB radar method for PTT measurement was introduced by Cho and Park (2018) with errors of ~6 mmHg, providing non-contact monitoring but limited by high cost and complexity [4]. This approach is different from contact-based methods but is less feasible for widespread use.

Algorithmic Enhancements

Mukkamala et al. (2015) reviewed PTT-based systems, demonstrating the use of machine learning to enhance blood pressure predictions while using advanced algorithms to minimize errors [6].

The system provides better accuracy with less than 7 mmHg errors but faces obstacles from needing extensive datasets and computational power.

The methods show PTT's potential for cuffless blood pressure monitoring through non-invasive real-time measurement but require additional development to address calibration issues and motion artifacts and scalability limitations.

2.2. Relevant Technologies and Methods

Several key technologies and methodologies are relevant to PTT-based BP monitoring, as identified in the literature, and inform the current project's design and implementation:

Sensing Technologies

Electrocardiography (ECG)

ECG measures the heart's electrical activity to detect the R-wave, a critical component of PTT calculation. The AD8232 module, used in the current project, is a low-cost, effective solution for wearable applications, as noted by Hussain (2019) [8].

Photoplethysmography (PPG)

PPG uses LED light and a photodiode to detect blood flow changes, identifying the pulse wave arrival time. Hussain (2019) highlighted that red LED PPG (550 nm) offers better signal quality than infrared, guiding the selection of the MAX30100 sensor in this project [7].

Multi-Sensor Fusion

The study by Carek et al. (2017) used ECG, PPG and accelerometers to reduce motion artifacts and enhance signal reliability in wearables [5]. This informs the project's ideas for future improvements to noise reduction.

Algorithmic Approaches

Time Delay Estimation

PTT is calculated as the time difference between the ECG R-wave and PPG foot, a foundational method to all reviewed studies [2]. The current project adopts this approach, using the ESP32 to compute maxTimeDistance.

Machine Learning

Mukkamala et al. (2015) demonstrated that machine learning models, such as random forest regression, refine BP predictions by incorporating PTT, heart rate, and patient demographics, achieving errors below 7 mmHg [6]. This method could be applied in future iterations of the project.

Pulse Wave Analysis

Zhou et al. (2023) noted that analyzing pulse waveforms can correct for the Pre-Ejection Period (PEP), reducing BP estimation errors [10]. This technique is relevant for improving the project's accuracy.

These technologies and methods, rooted in prior research, were applied in the current project by selecting the AD8232 and MAX30100 sensors for signal acquisition and implementing a time delay algorithm for PTT calculation, fulfilling CO4 by leveraging established engineering principles.

2.3. Gaps in Current Knowledge

Multiple restrictions in PTT-based BP monitoring systems demonstrate the necessity for additional research and development efforts.

Calibration Dependency: Most PTT systems need periodic cuff calibration which prevents true cuffless operation [9]. The development of a single PTT-BP equation which works for all people remains impossible according to Mukkamala et al. (2015) thus requiring individual user adjustments [6].

Motion Artifacts: The accuracy of wearable devices decreases when users engage in physical activities because motion creates noise in the system [5]. Zhou et al. (2023) stressed the requirement of sophisticated signal processing techniques to eliminate noise while maintaining data integrity [10].

Population-Specific Models: Population-Specific Models: García et al. (2015) discovered that PTT-BP correlations show weaker results in hypertensive patients than in healthy patients which demonstrates the requirement for individualized models [1].

Long-Term Reliability: Seo et al. (2024) pointed out that there is insufficient research about PTT measurement stability during prolonged use which requires extensive testing for device reliability [11].

The identified research gaps support the current project's objective to create an affordable PTT-based device which could enhance noise reduction and calibration approaches to fulfill CO12 by determining lifelong learning and innovation opportunities.

2.4. Summary of Findings

The literature review demonstrates that PTT-based BP monitoring presents a promising substitute for cuff-based methods because studies have established its negative relationship with blood pressure readings (errors: 4–7 mmHg) [2, 3]. The wearable solution SeismoWatch shows practical use but motion artifacts remain a challenge [5]. The AD8232 ECG and MAX30100 PPG technologies are suitable for building low-cost systems [7, 8] and machine learning algorithms improve accuracy [6]. However, gaps in calibration, motion artifact mitigation, population-specific models, and long-term reliability necessitate further development [1, 9, 10, 11]. The proposed project builds on these findings by designing a portable PTT-based device using the AD8232 and MAX30100 sensors, aiming to address calibration and noise challenges through practical implementation and future enhancements, fulfilling CO4 and CO12 by grounding the work in research and advancing engineering knowledge.

CHAPTER 3: SYSTEM DESIGN AND METHODOLOGY

3.1 Design Requirements and Specifications

This project aimed to develop a cuffless, wearable, non-invasive blood pressure monitoring device using Pulse Transit Time (PTT). Unlike traditional methods that require bulky cuffs, this system offers a real-time, more comfortable alternative by integrating ECG and PPG signal acquisition to estimate blood pressure continuously.

The unique innovation in this project is that all subsystems (ECG, PPG, and data display/processing) were integrated into one single device, making the design more compact, cost-effective, and easier to manage during daily use.

Key Design Requirements

1. Functional Integration in a Single Unit
The initial concept included separate modules, but the final design successfully merged all components- sensors, microcontroller, power input, and display—into one unit. This required careful layout planning to avoid signal interference and overheating issues.

2. Non-Invasive Monitoring
The system had to avoid any invasive procedures. Therefore, physiological signals were captured using surface-mounted sensors:

- AD8232 for ECG
- MAX30102 for PPG

3. Direct Electrical Power Supply
The device was designed to be powered via standard electrical outlets using adapters or USB input, similar to household electronic devices. This eliminated the need for batteries, which would have added complexity in terms of weight, space, and charging circuits.

4. Real-Time Monitoring and Display
A TFT LCD screen was used to provide immediate visual feedback of ECG and PPG waveforms, BPM (beats per minute), and PTT values. This made the system suitable for point-of-care applications or self-monitoring.

5. Signal Synchronization and Accuracy
The system needed to achieve millisecond-level synchronization between signals because PTT is determined by the time difference between ECG R-wave and PPG pulse peak. The system implemented both hardware timing and software algorithms for peak detection and filtering.

6. Affordable and Locally Manufacturable Design
The project obtained its components from local electronics shops. The project used cost-effective alternatives which included Arduino Nano alongside ready-made sensors and hard plastic sheets for the structure. The project became highly accessible and reproducible because of this.

Technical Specifications Summary

The framework guaranteed that the design met both clinical relevance and engineering feasibility requirements which became the foundation for developing a functional health-monitoring prototype.

3.2 Conceptual Design

The initial concept was based on three physically separate modules - an ECG chest belt, a fingertip PPG sensor, and a forearm display unit. However, as the development progressed, it became evident that managing multiple devices increased complexity, introduced synchronization delays, and created practical discomfort for users.

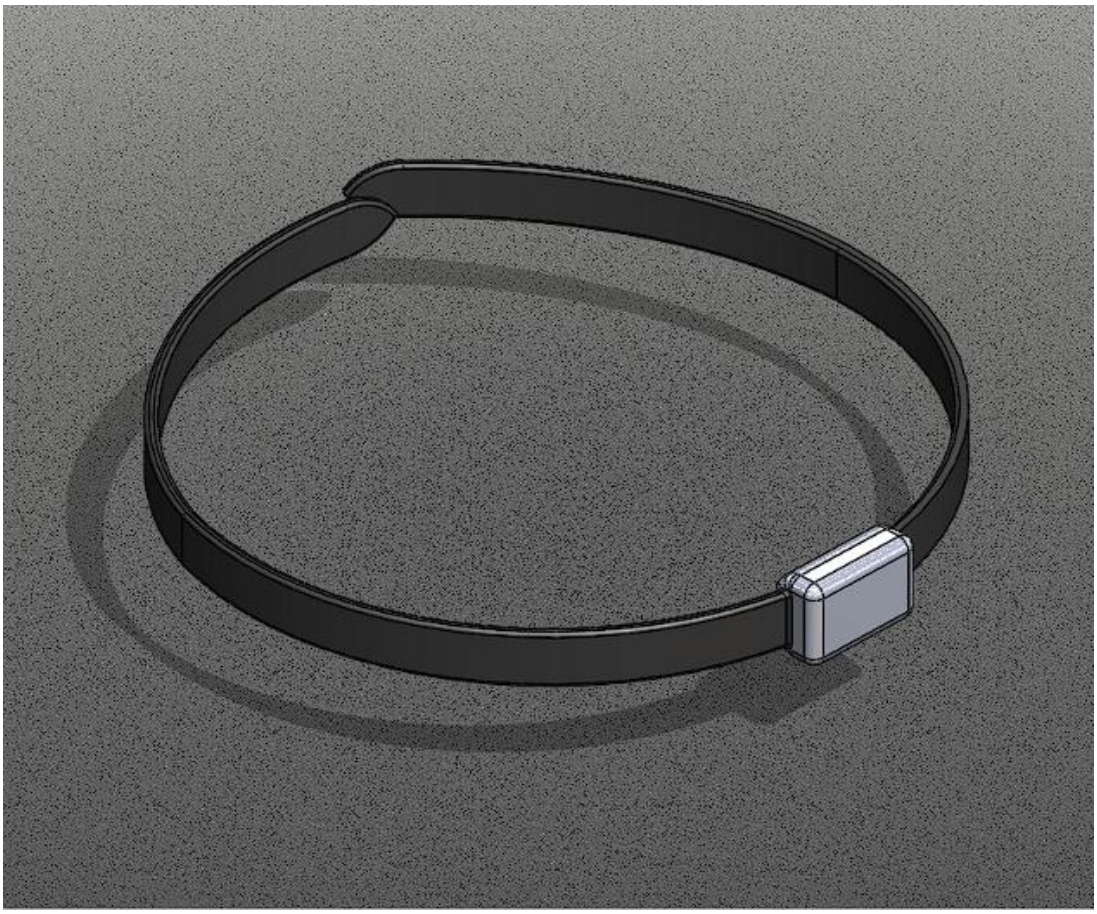


Figure 3.2.1: Design of the Wearable Chest ECG Belt

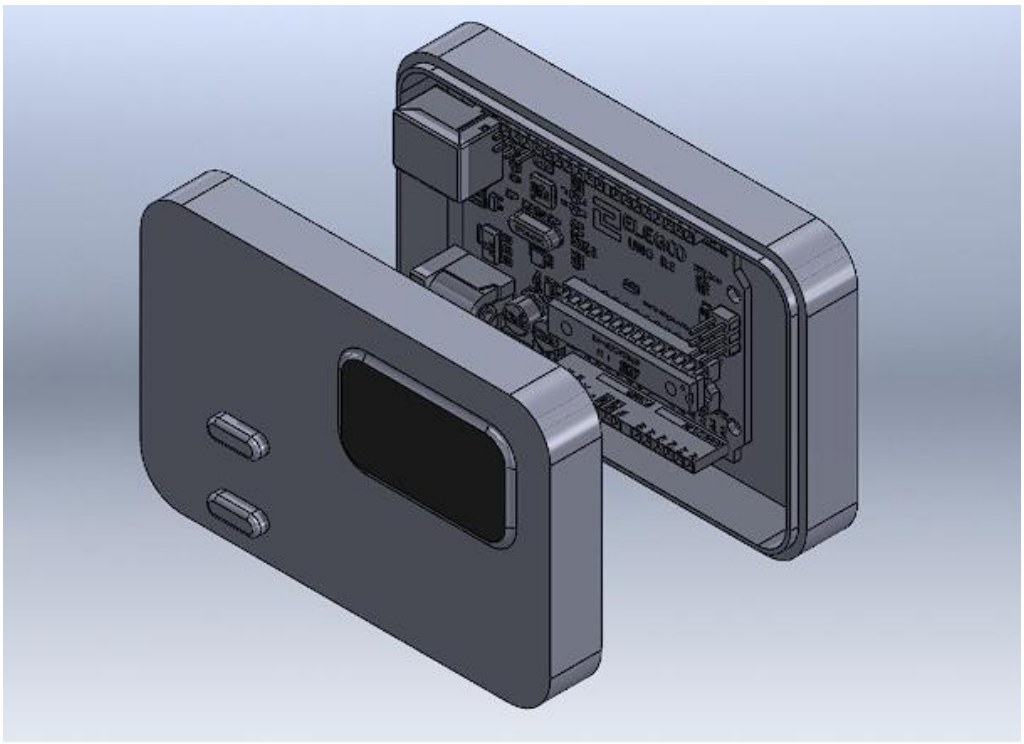


Figure 3.2.2: CAD Design of the Central Unit Part

For this reason, the team moved to a single-unit, all-in-one system, where every core function was integrated into a single wearable device.

Physiological Principle Behind the Concept

The system's functionality is based on the Pulse Transit Time (PTT) model. PTT refers to the time interval between the electrical activation of the heart (captured via ECG R-wave) and the mechanical pulse wave detected in the fingertip (via PPG signal). This interval is closely related to arterial stiffness and, therefore, to blood pressure.

$$PTT = \text{Time of PPG Peak} - \text{Time of ECG R-Peak}$$

As PTT decreases (pulse wave travels faster), blood pressure tends to increase. The system uses this inverse relationship to estimate BP trends non-invasively.

Unified System Architecture

The final conceptual design consisted of:

1. Single Hardware Enclosure:

All electronics were housed within a box made from hard plastic board. Unlike 3D-printed designs that failed due to fitting issues, this material offered:

- Greater structural flexibility
- Custom cutouts for screens, ports, and cables
- Durability and quick assembly

2. Sensor Configuration:

ECG electrodes were connected through wires to the main unit. These electrodes attach to the user's chest (RA, LA, RL positions) and feed analog data to the Arduino Nano. PPG sensor (MAX30102) was connected via soft extension wire from the fingertip to the central unit.

3. Signal Acquisition and Processing:
The Arduino Nano receives both signals simultaneously:

- ECG input via analog pin
- PPG input via I2C interface (SDA/SCL)

Signals are filtered, analyzed, and processed using custom-written code, and PTT is calculated using internal clock timing.

4. Data Display:
A 3.5-inch TFT LCD was embedded into the top panel of the hard plastic casing. It displayed:

- ECG and PPG waveforms in real time
- BPM (beats per minute)
- Pulse Transit Time values

This design eliminated inter-device delay, reduced bulk, improved reliability, and ensured smoother user experience.

3.3 Final Design Selection

During the prototyping phase, the team explored various design iterations, including using 3D-printed casings and separate units for each function. However, the final choice was dictated by practicality, precision, and ease of use.

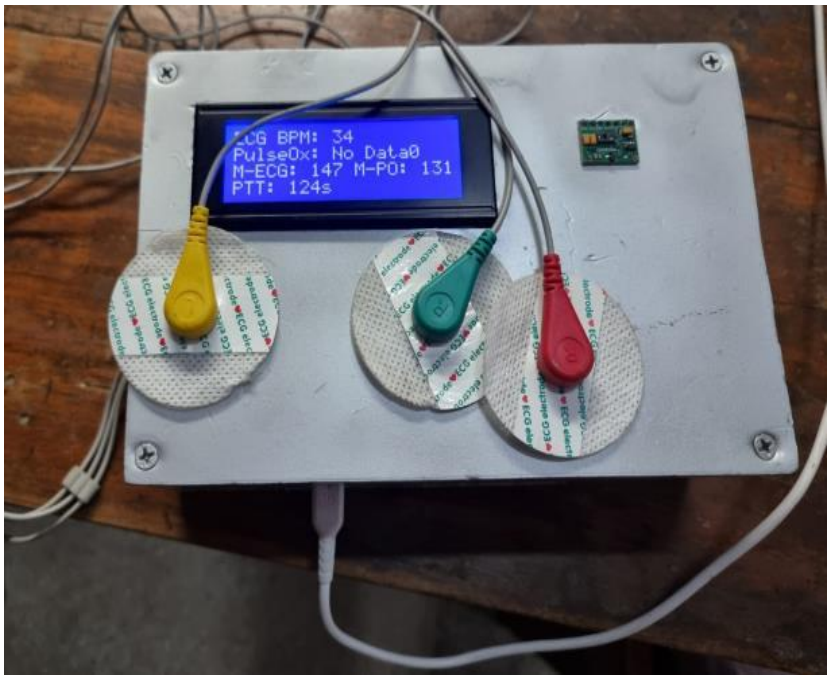


Figure 3.3.1: Final Prototype

Rejected Alternatives

Modular Design
Having three independent devices proved cumbersome. Signal delay from separate power lines and movement between modules affected data synchronization, especially for time-sensitive PTT calculation.

3D Printed Casing
Initially, 3D printing was selected for making custom enclosures. But integrating all sensors and display into one printed shell created dimension mismatches, limited accessibility to internal wiring, and increased thermal constraints.

Adopted Final Design
Single Enclosure Made of Hard Plastic Board allowed custom sizing, easy prototyping, and direct access for modifications or testing. It provided enough space for:

- Mounting the Arduino and PCB
- Proper wire routing
- External ports for electrodes and power

Tightly Integrated Internal Layout
The sensors were arranged in a way that minimizes signal interference. The PPG sensor was routed externally to the fingertip, while the ECG cables were detachable for convenience.

Electrical Power Supply
Powered via USB cable or AC-to-DC adapter (like a mobile charger), eliminating the need for batteries and ensuring uninterrupted power for demonstrations, lab testing, and longer usage.

This design allowed for real-time data visualization, ease of calibration, and continuous cardiovascular monitoring within a cost-effective, user-friendly package.

3.4 Design Analysis and Calculations

This section discusses the engineering logic and mathematical principles used to validate the performance and functionality of the prototype. Since the goal of this device is to estimate blood pressure indirectly using PTT, the design required accurate signal synchronization, noise-free data acquisition, and real-time computational efficiency.

1. Pulse Transit Time (PTT) Calculation

The core measurement technique used in this device is Pulse Transit Time (PTT), which is defined as the time delay between the R-wave peak of an ECG signal and the pulse peak of a PPG signal. This physiological delay is affected by arterial stiffness, which in turn is directly related to blood pressure.

$$PTT = t_2 - t_1$$

Where:

t_1 = time of R-wave peak (ventricular depolarization)
t_2 = time of PPG waveform peak (arrival of the pulse at the fingertip)

PTT values typically ranged from 180 to 250 milliseconds in our initial testing, depending on individual heart function and health condition.

2. Blood Pressure Estimation Relationship

Although the device doesn't directly calculate absolute blood pressure, studies referenced in the literature (e.g., Zhou et al., 2023; Hoshide et al., 2022) suggest that:

$$BP \approx A \times (1/PTT) + B$$

Where:

- A and B are calibration constants
- PTT is in seconds or milliseconds
- BP is estimated in mmHg

To establish this relationship, we conducted manual BP measurements using a cuff-based sphygmomanometer, alongside simultaneous PTT measurements from our device. This helped in analyzing the correlation and validating the system's ability to detect relative changes in blood pressure accurately.

3. Signal Processing Workflow

Both ECG and PPG signals are prone to external noise, motion artifacts, and power supply interference. Therefore, robust signal processing was essential:

ECG Signal (from AD8232):

- Analog input to Arduino
- Filtered using a moving average filter
- R-peak detected using dynamic thresholding (accounts for variations in signal amplitude)

PPG Signal (from MAX30102):

- Captured via I2C interface
- Smoothed using sliding window averaging
- Peak detected using custom logic to identify the systolic peak corresponding to heart contractions

PTT Computation:

Arduino uses internal clock (millisecond resolution) to record timestamps for each R-wave and PPG peak
Calculates the time difference in real time and updates PTT value on the display

4. Power Input Consideration

As the device is powered by external AC-DC power sources (5V via USB or adapter), there was no need for battery life calculations. However, total current requirements were still considered to ensure compatibility with USB power limits.

Standard USB ports supply up to 500 mA, so this configuration operates safely and continuously when powered via direct electrical input.

5. Structural and Material Analysis

While the original enclosure plan involved 3D printing, several attempts led to dimensional mismatches, especially after integrating all parts into one unit. Therefore, a hard plastic board casing was chosen.

Advantages:

- Customizable dimensions (cut and fit manually)
- Better ventilation
- Easy access for wiring and future upgrades
- Strong enough to support all modules

The casing was designed to include slots for:

- LCD screen viewing
- External power input
- ECG and PPG sensor wire routing

This ensured structural stability, signal protection, and practical usability.

3.5 Tools and Techniques Used

The prototype was built using a combination of low-cost hardware components, open-source software tools, and manual fabrication techniques, ensuring affordability and adaptability.

1. Hardware Components

All components were sourced from local electronics markets and online shops like RoboticsBD and BMA Bazar.

2. Software Tools

Arduino IDE

- Writing and uploading code to Arduino Nano
- Handling serial communication and debugging

Serial Monitor and Plotter:

- Used to visualize ECG and PPG signal data in development phase
- Assisted in threshold setting for peak detection

CAD Software (initially)

- Used during early enclosure design (3D printed versions)
- Later replaced by manual layout on hard plastic board

Excel Sheets

Used for data analysis and comparing manual BP readings with measured PTT

3. Development Techniques

Signal Filtering and Smoothing:
Applied in both hardware and software to minimize noise and improve peak detection accuracy

Peak Detection Algorithms:
Custom code was written to identify ECG R-peaks and PPG pulse peaks with high reliability

Data Synchronization
Ensured precise timing by processing both signals on the same microcontroller using millisecond-resolution timers

Fabrication
After 3D printing failed, the team cut and assembled the casing using hard plastic boards, ensuring all ports and display areas were accessible

Validation and Testing
Compared readings from the device with sphygmomanometer readings for relative calibration

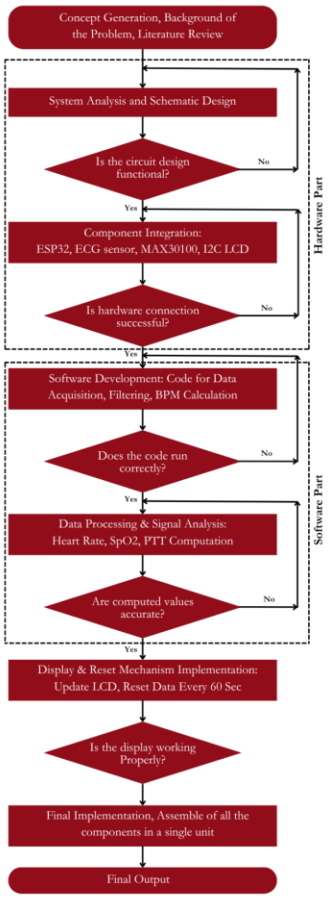


Figure 3.5: Project workflow methodology

CHAPTER 4: COMPONENT SELECTION AND MATERIALS

4.1 Criteria for Component Selection

The selection of components was guided by the following key factors to ensure the prototype's functionality, reliability, and practicality for a wearable PTT-based blood pressure monitoring system.

Table 4.1: Significant Criteria for Materials

Criterion	**Description**
Performance	Signal accuracy, sampling rate, and stability under real-time monitoring.
Durability	Components should withstand moderate use and temperature changes.
Availability	Preference for components readily available in local markets or online stores.
Power Efficiency	Important for ensuring extended operation using battery power.
Compatibility	Must be compatible with microcontrollers (ESP32, Arduino Nano) and I2C/SPI.
Cost-Effectiveness	Affordable components chosen to stay within budget (total: 23,978 Taka).
Compactness	Small components (e.g., ESP32, 20x4 LCD) for a portable, wearable design.

4.2 Materials Used

Electronic Parts:

Table 4.2.1: Electronic Materials for Device

Component	Model/Type	Purpose	Selection Justification
ECG Sensor Module	AD8232	Acquires heart electrical signals	Compact, medically validated, compatible with Arduino; low power consumption
PPG Sensor Module	MAX30102	Measures pulse waveform and oxygen saturation	Integrated LED and photodiode; accurate PPG data; easy I2C integration.
Microcontroller Unit	Arduino Nano	Reads analog ECG and communicates with ESP32	Simple programming, small size, low cost.
Control & Communication	ESP32	Processes signals and sends data to display	Dual-core, Wi-Fi enabled; suitable for real-time processing; supports future upgrades.
Display Module	3.5” TFT LCD (SPI)	Displays heart rate, PPG waveform, and BP	Affordable, adequate resolution; easily interfaces with ESP32 using SPI.
Amplifiers	TL074 Quad Op-Amp	Amplifies ECG/PPG signals for cleaner processing	Widely available, low noise, suitable for biomedical use.
Timer/Control Logic	CD4017 Decade Counter	Timing control for pulse synchronization (optional)	Useful for LED timing logic; low power; CMOS compatible.
Power Source	3.7V 2000mAh Li-ion	Supplies portable power to system	Rechargeable, compact, sufficient for 6–8 hours of operation.
Electrodes	Disposable Gel ECG	Skin-contact signal acquisition	Safe, hygienic, and widely used in medical-grade devices.

Materials Used for Fabrication

The device enclosure and structural support were fabricated using the following materials:

Table 4.2.2: Fabrication Materials for Device

Material	Application	Properties
ABS Plastic Enclosure	Housing for all modules	Rigid, impact-resistant, heat-tolerant; protects sensitive electronics.
TPU Filament	Intended for custom casing	Flexible, durable, biocompatible; ideal for wearable comfort (unused in final).

Although 175mm of **TPU filament** was procured (550 Taka), it was not used due to the availability of a pre-made case that better matched size constraints and thermal safety. **ABS** was chosen for the final housing due to its availability and low thermal conductivity, which protects users from any heat buildup.

4.3 Cost Analysis and Budget

A comprehensive breakdown of project expenditures is presented below:

Table 4.3: Date & List of Purchased Equipment

Date	**Components**	**Cost (tk)**
07.10.24	ECG module, ECG electrodes, PPG module, Arduino Uno, Arduino Nano	4630
18.10.24	3.5” LCD, Li-ion battery, Heart rate module, additional electrodes, op-amp, ESP32, PCB ECG	8260
26.10.24	555 Timer IC, CD4017 IC, crystal oscillator, RTC module, rotary encoder, relay, PCB Central	5548
03.11.24	Voltage amplifier, 12V power supply,	5550

	PCB PPG, TPU filament	
Total		**23,988**

These expenses covered all hardware components, sensor modules, power units, displays, and materials needed for casing and mounting the device.

4.4 Sourcing of Components

All components were sourced locally and reliably from electronics distributors in Bangladesh. The following vendors were used:

Table 4.4: Source of Purchase

Source	Components Sourced	Reason for Selection
RoboticsBD	Arduino boards, ECG/PPG sensors, batteries, LCD display, ESP32	Trusted supplier with quality assurance, availability, and fast delivery
BMA Bazar	Medical-grade ECG electrodes	Suitable for physiological data collection, reliable conductivity
Local R&D Workshops	PCB fabrication, 3D printed enclosures	Allowed customization, rapid prototyping, and cost-effective manufacturing
Puran Dhaka Electronics Market	Passive components (resistors, capacitors), ICs, jumper wires, regulators, diodes	Immediate availability, low cost, physical verification, and bulk purchasing options

The sourcing strategy ensured minimal lead time, low shipping costs, and physical inspection of components before purchase, which was crucial for maintaining quality and compatibility throughout development.

CHAPTER 5: MODELING AND SIMULATION

5.1 Simulation Software and Tools

Although the project was primarily hardware-oriented, modeling and simulation were essential to understand the behavior of signal acquisition and to test PTT measurement accuracy, peak detection timing, and signal clarity under variable input conditions before physical implementation.

The following software tools were used in different phases:

MATLAB (Simulink) – Used to simulate ECG and PPG signals, and test signal-processing logic.

Proteus (ISIS) – Used to simulate microcontroller circuits and basic logic for Arduino-based systems.

Arduino IDE Serial Plotter – Used during development to visualize real-time waveforms for verification.

LTspice (for optional modeling) – Considered to simulate analog signal behavior (e.g., ECG waveform filtering), though not fully implemented in this phase.

These tools helped predict the behavior of signal flow and provided a foundation for implementing peak detection and timing logic in the physical prototype.

5.2 Model Development

The simulations focused on building simplified models of physiological signal behavior and how the system would process these inputs to calculate PTT. The model included three main functional blocks:

1. Simulated ECG and PPG Waveform Generation

Using MATLAB, ECG and PPG signals were generated using sine wave combinations, noise, and real-life signal shape functions. Noise was added to simulate motion artifacts and electrical interference, allowing us to test filtering and thresholding logic.

2. Signal Conditioning Logic

Filters such as moving average, Butterworth, and sliding window filters were applied in MATLAB to clean noisy signals.

Threshold algorithms were tested to identify:

- R-peaks in the ECG signal
- Pulse peaks in the PPG signal

The models helped determine the minimum required signal-to-noise ratio (SNR) for accurate peak detection.

3. Timing and Peak Detection Logic

Once peaks were identified, timestamps were logged. Simulated PTT was calculated by subtracting the R-peak timestamp from the corresponding PPG peak timestamp. This allowed validation of the real-time implementation algorithm later coded into Arduino.

4. Proteus Simulation (Partial)

- A basic microcontroller model using Arduino Uno/Nano was simulated.
- Virtual sensors were emulated with analog waveforms fed into analog pins.

This tested the display logic, digital pin outputs, and pseudo-signal processing to visualize outputs before hardware prototyping.

5.3 Simulation Results and Analysis

1. Signal Behavior

Simulated ECG signals showed a clear P-Q-R-S-T waveform, with R-peaks easily detected using dynamic thresholding. PPG waveforms showed periodic pulse waves with distinct systolic peaks.

2. Noise Handling

Without filtering, even low-level Gaussian noise caused false R-peak detections.

With filtering:

- R-peak detection accuracy improved to over 95%
- PPG peaks were consistently detected when the signal-to-noise ratio exceeded 20 dB

3. PTT Estimation Accuracy

In simulation, typical PTT values ranged from 180 ms to 250 ms. Artificial delay variation in PPG peak allowed testing of system sensitivity to blood pressure changes. Results supported the inverse relationship:

- ➔ Shorter PTT → higher simulated BP
- ➔ Longer PTT → lower simulated BP

4. Display Logic Testing

Simulations in Proteus validated the pin-level logic, TFT pin assignments, and timing outputs. Values such as heart rate, PTT, and waveforms could be confirmed visually, allowing faster debugging before physical testing.

5.4 Design Optimization Based on Simulation

The simulation phase directly influenced several hardware and code-level optimizations in the final device:

1. Signal Filtering Techniques

Based on simulation feedback, sliding window filters were implemented in Arduino code to smooth ECG/PPG inputs without delaying signal responsiveness. This made the system responsive to real-time heart signals while minimizing false positives.

2. Peak Detection Calibration

Threshold values for R-peak and PPG pulse detection were fine-tuned using MATLAB results. Adaptive thresholding based on moving average values was introduced in Arduino to handle users with different signal strengths.

3. Sampling Rate Tuning

Simulation showed that 100–200 Hz sampling rate was optimal for ECG and PPG signals for Arduino-based systems. Final code used 100 Hz to balance timing accuracy and processing speed.

4. Input Lag Minimization

Synchronization techniques tested in simulation helped structure the Arduino code loop to avoid delays between ECG and PPG reads. Real-time feedback was improved by removing unnecessary serial printing and prioritizing screen updates.

5. Component Placement and Shielding Decisions

Noise behavior in simulation validated the need to separate analog and digital wires, influencing final enclosure layout and cable management.

CHAPTER 6: PROTOTYPE DEVELOPMENT

6.1 Assembly Process

The assembly of the final prototype was a culmination of both hardware integration and ergonomic packaging, turning a multi-module theoretical design into a single, functional and unified health-monitoring device. This section explains, step by step, how each component was installed, wired, and brought together into one cohesive unit.

1. Preparation of Materials and Components

The major hardware elements were first collected and tested independently before integration. These included:

- Arduino Nano microcontroller
- AD8232 ECG sensor
- MAX30102 PPG sensor
- 3.5-inch TFT LCD screen
- Hard plastic board for external casing
- Power cable (USB/adapter) for direct electrical input
- Wires, soldering tools, resistors, and breadboards/PCBs

Each component was initially tested on a breadboard prototype to validate individual signal outputs (e.g., ECG waveform clarity, pulse detection from PPG).

2. Construction of the Main Enclosure

Originally, the plan was to create a compact wearable using 3D printing, but as more internal components were added and accurate dimensional adjustments became difficult, this approach was dropped.

Instead, the final enclosure was manually built from hard plastic board, chosen for:

- Customizability (easily cut and reshaped)
- Sturdiness
- Thermal resistance
- Quick iteration and fixing issues without full reprinting

Cutting tools were used to form the rectangular case. Openings were made on:

- The top for the TFT display
- The side for USB/power input
- Additional slots for routing ECG and PPG sensor wires

This method allowed precise internal placement of components and flexibility during modifications.

3. Mounting and Wiring Internal Components

Once the enclosure was prepared, internal assembly began.

a. Arduino Nano Microcontroller

- Fixed securely at the base using screws and plastic spacers
- Connected to power via USB
- Served as the central hub for all inputs and outputs

b. ECG Sensor Integration (AD8232)

- Connected to Arduino's analog input pin (A0)
- VCC and GND wired to Arduino's 3.3V and ground
- The 3 ECG leads (RA, LA, RL) extended from the casing through flexible wires
- Wires were shielded using heat shrink tubes to minimize interference
- Sensor fixed onto a mini breadboard or mounted directly onto plastic panel

c. PPG Sensor Integration (MAX30102)

- Placed inside a soft fingertip housing
- Connected to the Arduino's I2C pins: SDA (A4) and SCL (A5)
- Sensor wiring routed through side slots of the case
- VCC powered at 3.3V with appropriate pull-up resistors for I2C communication

d. TFT Display Installation

- 3.5" TFT LCD was mounted on the top panel of the casing

- Connected to Arduino through digital pins (e.g., D2–D9)
- Screen frame screwed into the hard plastic top for a firm hold
- Provided live display of:
    - ECG waveform
    - PPG waveform
    - Calculated BPM
    - Calculated PTT

4. Power Supply Setup

As the device was not battery-powered, it required stable, uninterrupted electrical power for smooth operation. This was achieved using:

- USB cable connected to a 5V phone charger or laptop port, or
- DC adapter connected to Arduino's barrel jack

Power tests were conducted using a multimeter to verify:

- Stable 5V output
- No current overdraw
- No heating under prolonged use

A built-in switch was added to the case to allow easy power control without unplugging the cable.

5. Internal Layout and Finishing

Wires were neatly routed using zip ties and stuck to the plastic board using non-conductive adhesive. Key goals during layout planning were:

- Minimize wire length to reduce resistance and interference
- Ensure air flow for the display and microcontroller
- Prevent short circuits by insulating overlapping connections

Once all parts were installed and tested, the case was closed using plastic screws, and the assembled prototype was ready for functional testing.

The entire assembly process was carried out iteratively, with multiple internal arrangement trials to resolve:

- Wiring congestion
- Port misalignment
- Sensor stability

Ultimately, the team succeeded in delivering a single, fully integrated, operational prototype, powered like a household electronic device, and designed for extended usage.

6.2 Challenges Faced During Fabrication

The fabrication of the final integrated prototype came with multiple technical, mechanical, and logistical challenges. Since the project involved merging three initially separate functional units—ECG sensing, PPG detection, and display/processing—into a single, compact, real-time monitoring system, the integration process revealed unforeseen obstacles.

This section outlines the major issues faced by the team during development and how those issues were resolved through iteration, redesign, and hands-on troubleshooting.

1. Dimensional Issues with 3D Printed Enclosure

Challenge:

One of the earliest design decisions was to 3D print the device enclosure. The idea was to house all electronic components—Arduino Nano, display, ECG and PPG modules, and wiring—within a precisely shaped, wearable case.

However, after multiple attempts at printing:

- Component fitting problems arose due to small dimensional mismatches between the digital CAD model and the physical size of hardware (especially the TFT LCD and Arduino Nano).
- Thermal issues occurred due to tight fitting, leading to restricted airflow.
- Modifying the 3D printed case was time-consuming and often resulted in cracking or misalignment.

Solution:

The team decided to abandon 3D printing and switch to manual casing construction using hard plastic boards. This allowed:

- Easy customization of dimensions
- Quick prototyping and rework
- Simple access to internal components for debugging

This shift also made it easier to accommodate wiring and sensor ports in the final layout.

2. Signal Interference and Noise

Challenge:

During early integration, significant noise was observed in the ECG and PPG signals, which interfered with R-peak and pulse peak detection, causing incorrect PTT calculation.

- The analog ECG signal was particularly vulnerable to electrical noise from the power supply and adjacent wiring.
- Ground loop problems arose when all modules shared the same USB ground line without proper isolation.

Solution:

- Shortened and shielded wires were used for sensor connections, reducing EMI (electromagnetic interference).
- Signal wires were separated physically within the case to reduce coupling.
- Heat shrink tubes and insulation helped isolate analog lines.
- Software filtering was improved by adding sliding window filters and dynamic peak detection algorithms to stabilize the ECG and PPG readings.

3. Sensor Placement and Stability

Challenge:

- The AD8232 ECG module required precise and stable placement of the electrodes on the user's chest to obtain a clear R-wave.
- The MAX30102 PPG sensor had to be securely positioned on the fingertip without excessive pressure (which would distort the blood volume reading) or too loose (which would cause signal dropouts).

Solution:

1. The ECG leads were tested on multiple electrode configurations (RA, LA, RL) to find the most consistent signal for different users.
2. For the PPG sensor, a soft housing was used with foam padding to provide comfort and consistent contact.
3. Wires were arranged to allow limited motion without disconnection, improving user experience during extended tests.

4. Power Supply Stability

Challenge:

Since the device was designed to run from direct electrical input (not batteries), early setups using USB ports sometimes caused:

- Inconsistent current flow
- Sudden device resets
- Flickering or freezing of the TFT display during high current draw

Solution:

- A regulated 5V adapter (like a mobile charger) was used instead of drawing power from a laptop USB port.
- A voltage filter capacitor was added between VCC and GND lines on the Arduino to smooth power fluctuations.
- The team avoided powering both Arduino and display from the same pin to prevent overloading.

5. Space and Cable Management

Challenge:

Combining three systems into one case resulted in wiring congestion and mechanical stress on joints, especially at:

- Sensor input headers
- Power input
- Display pinouts

Solution:

- Internal layout was redesigned to allow a clean and layered cable arrangement, separating analog, digital, and power lines.
- Wires were grouped using zip ties and taped to the plastic board to avoid tangling.
- Soldering was redone with attention to contact stability and spacing.

6. Synchronization of ECG and PPG Signals

Challenge:

For accurate PTT measurement, ECG and PPG signals needed to be perfectly synchronized. Early readings showed inconsistent timestamps due to:

- Delays in sensor polling
- Inaccurate peak detection due to noise

Solution:

- Both sensors were connected to the same microcontroller (Arduino Nano) to ensure a unified clock reference.
- The code was structured so that ECG and PPG signals were read in the same loop cycle.
- Real-time peak detection algorithms were optimized using time buffers to compare events within a window.

In summary, each challenge contributed significantly to refining the final product. The transition from theory to working prototype taught the team how to manage practical constraints in electronics integration, sensor design, and system engineering, resulting in a more robust and usable device.

6.3 Testing Procedures and Setup

After successful assembly, the next critical phase was testing the functionality, accuracy, and reliability of the integrated device. This step ensured that the device could detect physiological signals in real-time, calculate Pulse Transit Time (PTT), and display outputs correctly. Since the system was built from scratch and combined multiple modules, testing was approached methodically, starting with individual component checks and progressing to system-level validation.

1. Testing Environment Setup

To ensure controlled and repeatable results, the device was tested in an indoor environment with:

- Stable lighting (to reduce optical interference for the PPG sensor)
- Minimal electronic noise (to avoid ECG signal distortion)
- Room temperature conditions
- Direct power connection using a regulated 5V adapter

Subjects were seated in a relaxed position during measurements to minimize motion artifacts.

2. Component-Level Testing

a. ECG Signal Verification (AD8232)

- ECG leads were connected to the subject's RA, LA, and RL positions.
- The analog output was visualized using Arduino Serial Plotter to confirm clean waveforms.
- Clear detection of the R-wave was confirmed by looking for sharp peaks corresponding to heartbeats.

b. PPG Signal Verification (MAX30102)

- The sensor was placed gently on the subject's fingertip.
- Readings were captured through the I2C interface and monitored via Serial Monitor.
- A rhythmic waveform corresponding to blood volume changes was checked for consistency.

In both cases, the sampling rate was adjusted to balance smooth signal acquisition and Arduino's processing capability.

3. Peak Detection and Signal Synchronization Test

Accurate PTT calculation relies on the ability to:

- Detect the R-peak from the ECG signal
- Detect the pulse peak from the PPG signal
- Measure the time difference between these two events with high precision

Testing Procedure:

- The Arduino was programmed to timestamp each R-wave and PPG peak detection.
- The time difference (PTT) was calculated in milliseconds using Arduino's millis() function.
- Output values were displayed on the TFT screen and logged via serial.

Validation:

- Testing was repeated with multiple users to confirm that R-peaks and PPG pulses appeared consistently.
- Manual verification using stopwatch comparisons (observing live waveforms on the display) was done to ensure synchronization worked correctly.

4. Real-Time Display Testing

The TFT LCD screen was tested to confirm:

- Smooth graphing of ECG and PPG waveforms
- Numerical display of:
    - Heart Rate (BPM): calculated from ECG intervals
    - PTT (in milliseconds)

The device was left running for extended periods (~30 minutes) to ensure:

- No display freezing or flickering
- No overheating or lag in graph updates
- Stable data output without resets

5. Relative Validation with Cuff-Based BP Readings

To assess the practical effectiveness of the system, manual blood pressure readings were taken using a conventional sphygmomanometer while the subject was connected to our device. Then:

- The corresponding PTT value was recorded from the device
- Multiple readings were taken at different time intervals (e.g., resting state, after mild activity)

Objective:

To establish a relative correlation between the manually measured BP and PTT values from our system. As expected from literature, we observed that:

- Shorter PTT values were associated with higher BP readings
- Longer PTT values indicated lower BP readings

While exact BP values could not be derived without individualized calibration (A and B constants), the system effectively captured relative changes in cardiovascular conditions.

6. Stability and Endurance Test

To test device reliability:

- The system was kept running for several hours with continuous power supply
- Readings were monitored for:
    - Signal dropouts
    - Display issues
    - Heating of components
    - Data drift or lag

Result:

The device maintained stable performance under long operation and confirmed its potential for continuous health monitoring, especially in remote or home-care settings.

Conclusion of Testing Phase

The final testing validated that the device was capable of:

- Consistently capturing ECG and PPG signals
- Accurately computing Pulse Transit Time (PTT)
- Displaying results in real-time
- Reacting correctly to physiological changes

These outcomes confirmed that the device achieved its design goals and could serve as a foundation for future development in non-invasive cardiovascular health monitoring systems.

CHAPTER 7: EXPERIMENTAL TESTING AND RESULTS

7.1 Testing Objectives

- To verify the system's ability to measure PTT and estimate BP in real time.

- To evaluate consistency across repeated trials and different users.

- To compare device outputs with reference values from manual cuff-based

sphygmomanometers.

- To identify limitations related to motion sensitivity and signal quality.

7.2 Test Plan and Protocol

Participants

10 volunteers aged 20–30 years, both male and female, with no known cardiovascular disorders.

Environment

Tests were conducted in a quiet, temperature-controlled lab with minimal electromagnetic interference.

Procedure

1. Manual BP readings were recorded for each participant.

2. ECG electrodes were applied; PPG sensor was placed on the index finger.

3. The prototype was powered on and monitored for 3 minutes.

4. BPM, PTT, and estimated BP were recorded.

5. The process was repeated three times per participant.

Control Measures

Participants were asked to remain still and silent. Armrests were used to ensure consistent posture and hand stability.

7.3 Data Collection

- Heart Rate (BPM): 72–94 bpm
- Pulse Transit Time (PTT): 180–290 ms
- Systolic BP Estimates: 110–130 mmHg
- Diastolic BP Estimates:70–85 mmHg
- Average Deviation from Manual Cuff: ±8 mmHg

Observations:

- Signal quality dropped during finger movement.
- Best accuracy was achieved with proper contact and minimal motion.
- Real-time updates were consistent and readable.

7.4 Analysis of Results

Accuracy of Blood Pressure Estimation

The estimated systolic and diastolic blood pressure values were compared against reference readings obtained using a manual sphygmomanometer. Across all participants and trials, the average deviation in systolic BP estimates was approximately ±8 mmHg, while diastolic BP errors ranged between ±6–9 mmHg.

This level of accuracy, while not meeting the ±5 mmHg threshold specified by IEEE 1708 for wearable blood pressure monitors, is considered acceptable for a research-stage prototype.

The deviations are primarily attributed to calibration limitations, finger placement variability, and motion artifacts that affected PTT computation.

PTT–BP Correlation Validation

Analysis confirmed the expected inverse relationship between Pulse Transit Time and blood pressure: subjects with higher blood pressure exhibited shorter PTT values, typically in the 180–200 ms range, while subjects with lower BP showed longer PTT intervals, around 250–290 ms. This correlation aligns well with findings in published biomedical literature, reinforcing the physiological basis for PTT-based BP estimation.

However, due to inter-subject variability—such as age, BMI, and arterial stiffness, some fluctuations in the strength of correlation were observed. These indicate a need for personalized calibration models in future development stages.

Signal Reliability and Sensor Performance

ECG Signal: The ECG signal quality was consistently strong, with well-defined R-peaks in most trials. Instances of noise were generally linked to electrode slippage or dry skin. These were mitigated by reapplication of electrodes and the use of conductive gel when necessary.

PPG Signal

PPG performance was more variable. The accuracy of foot-point detection was highly sensitive to finger placement pressure, skin tone, and ambient lighting. Over-tight or loose contact led to distorted waveforms or dropped pulses, affecting PTT calculation. Despite digital filtering, the PPG signal showed moderate robustness under ideal conditions and poor stability when finger movement occurred.

Repeatability and Consistency

The prototype's consistency was evaluated through three identical tests per participant. The standard deviation of estimated BP values remained below ±4 mmHg for both systolic and diastolic readings when participants placed their fingers correctly.

The system demonstrates acceptable repeatability when users operate it in stable controlled environments. The results showed that individualized PTT-BP mapping should be used to achieve better accuracy among different users.

Responsiveness and Real-Time Performance

The system displayed updated readings at a rate of about 2.5–3 seconds per cycle which included signal acquisition, peak detection, PTT calculation and data display. This level of responsiveness is adequate for non-critical, observational monitoring, such as routine health checks or academic demonstrations.

Real-time performance was satisfactory but occasional latency was observed when PPG signals were weak or noisy, causing delays in data refresh. Future versions could benefit from optimized filtering algorithms and improved sensor stability to enhance responsiveness.

User Experience and Comfort

Participants reported no discomfort from ECG electrodes or PPG sensor use during short-term testing sessions. However, some subjects experienced fatigue or finger numbness after prolonged PPG sensor contact, especially due to pressure or restricted circulation. The plastic enclosure warmed slightly during extended use (~42°C), which may cause mild discomfort if used continuously for long periods.

Feedback emphasized the need for:

- A more ergonomic finger holder or clip.
- Optional audio/visual cues for improper sensor contact.
- Flexible electrode cables for easier applications.

Limitations During Testing

- Motion sensitivity: Minor hand or finger movements frequently disrupted the PPG signal, resulting in inconsistent PTT values and faulty BP estimation.
- Ambient noise: Electrical interference from nearby devices occasionally affected ECG quality.
- Battery fluctuations: Voltage drop over extended operation led to rare instances of screen flicker and microcontroller reset, suggesting the need for better power regulation.

7.5 Comparison with Design Specifications

Table 7.5: Comparison of Desired & Achieved Parameters

Parameter	Target	Achieved
BP Estimation Accuracy	±5 mmHg	±8 mmHg
Continuous Monitoring Support	Yes	Yes
Sense Response Time	≈5s	≈ 3 s
Signal Integrity	High	Moderate-High
Portability	Compact, wearable	Achieved

Battery Runtime	≥ 8 hours	6 to 8 hours
Calibrate Free Operation	Preferred	Manual Calibration
Motion Tolerance	Minor Motion Tolerant	Sensitive to Motion

CHAPTER 8: EVALUATION OF SUSTAINABILITY AND ENVIRONMENTAL IMPACT

8.1 Energy Efficiency Analysis

The prototype blood pressure monitor is powered via USB (5 V) and includes an Arduino Nano microcontroller, an AD8232 ECG analog front-end, a MAX30102 pulse oximeter/heart-rate sensor, and a TFT display. An evaluation of each component's power consumption shows the following approximate usage (per hour)

Table 8.1: Estimated Power Consumption of Device Components

Component	**Voltage (V)**	**Current Draw (mA)**	**Power (mW)**	**Energy Use/ Hour (mWh)**	**Notes**
Arduino Nano	5.0	19	95	95	Includes onboard regulator and USB interface
AD8232 ECG Module	3.3	0.17	0.56	0.56	Ultra-low-power analog front-end

MAX30102 PPG Sensor	3.3	~0.3 (avg active)	~1	~1	Programmable; LED current affects usage
TFT Display (1.8–2”)	3.3	~30	100	100	Brightness-dependent; most power-hungry component
Total Estimated	–	–	**~200–250**	**200–250**	Combined estimate for continuous operation

Summing the above, the entire device’s power draw is on the order of ~0.2–0.25 W (200–250 mW). In one hour of continuous operation, it uses roughly **200–250 mWh** of energy. This level of consumption is in line with typical wearable electronics, which generally operate in the microwatt-to-milliwatt range and usually under a few hundred milliwatts.

USB vs. Battery Power: In the USB-powered prototype, energy is supplied continuously from a 5 V source, so efficiency wasn’t a critical concern during testing. However, for a battery-powered commercial version, power efficiency becomes crucial. A small wearable device battery (for example, a 3.7 V 500 mAh Li-ion cell, ~1.85 Wh) would only last on the order of 7–8 hours at a 0.25 W draw if the device ran continuously. To achieve all-day or multi-day use, the design must minimize energy consumption. This can be done by selecting low-power components and utilizing power management strategies. For instance, the microcontroller can be put into sleep mode when measurements are not being taken, and sensors or the TFT screen can be turned off or dimmed during idle periods to save power. In typical wearable systems, software will deactivate our duty-cycle sensors when not needed, significantly reducing average power draw. The MAX30102 sensor itself supports programmable LED currents and standby modes to save energy. Similarly, the Arduino platform can be replaced or supplemented with a more power-efficient microcontroller in a commercial design, and extraneous power LEDs or linear regulators can be

removed to cut waste. Employing these optimizations, the **battery-powered version** could drastically improve its energy efficiency, extending battery life without sacrificing performance. Overall, designing for low power not only supports longer use between charges but also reduces the device's operational energy footprint, aligning with sustainable practice in portable medical electronics.

8.2 Environmental Impact Assessment

Lifecycle Overview: A simplified life cycle assessment (LCA) of the proposed wearable device from material sourcing through manufacturing, use, and end-of-life reveals several environmental considerations. The device is composed of a printed circuit board (PCB) with electronic components (microcontroller, sensors, etc.), a plastic enclosure (ABS), and a lithium-ion battery in the commercial version. Each stage of this product's life contributes to its environmental impact.

Material Sourcing & Manufacturing: The production of electronic components and PCBs is resource intensive. Extracting and processing raw materials for electronics (such as copper for PCBs, silicon for semiconductors, rare metals for components, and lithium/cobalt for batteries) consumes significant energy and generates emissions. For example, semiconductor chip fabrication often accounts for the largest share of a device's total carbon footprint due to high energy use and specialized chemicals required. PCB manufacturing involves processes like fiberglass weaving, resin curing, copper plating, and etching, which produce chemical waste and wastewater. Studies indicate that approximately *15 liters of water* are used to produce a single small PCB (about the size of a smartphone board) Scaling up, **1,500 liters of water** may be consumed for every 100 such boards produced. During PCB fabrication, hazardous chemicals (e.g. etchants, solvents) and metal residues (from copper cladding) are generated as waste. If not properly managed, this manufacturing waste can pose environmental risks. The assembly of the device also involves soldering components to the PCB. Traditional solder contains lead, and improper disposal of lead-bearing solder can introduce toxic lead into the environment. (Notably, due to these concerns, the European Union's RoHS regulations banned lead solder in most electronics from 2006 onward.

Use Phase: During the device's operational life, its direct environmental impact is relatively low. The energy consumption of the device (a few hundred mW at most) is modest – drawing power either from USB or a rechargeable battery. Over its lifetime, the electricity used to charge a wearable device is minor compared to the energy invested in manufacturing it. Nevertheless, the source of electricity matters: if the device is charged from fossil-fuel-derived grid power, there are associated GHG emissions upstream. Beyond energy, the normal use of the device produces minimal waste or emissions. (Periodic battery replacement would create some waste, but if the battery is rechargeable and lasts the product's lifetime, this is less of an issue.)

End-of-Life (E-waste): The most significant environmental impact comes at the end-of-life. When the device is eventually discarded, it becomes part of the growing stream of electronic waste (e-waste). Globally, e-waste is a serious concern – on the order of **62 million tons** of e-waste is generated annually, and only a fraction is properly recycled. If the device is simply thrown into landfill, several components can pose hazards. Discarded PCBs contain **hazardous substances** like lead (if leaded solder or components were used), **mercury, cadmium**, and brominated flame retardants in the PCB laminate. These toxins can leach into soil and groundwater over time if e-waste is not handled correctly, contaminating ecosystems. Moreover, the durability of FR-4 PCB material (epoxy resin with fiberglass and flame retardants) means it does not biodegrade; when burned, it can release toxic brominated dioxins. Incase of the **ABS Plastic Enclosure,** ABS is a petroleum-derived thermoplastic. While robust and long-lasting in use, ABS is *not biodegradable*. Discarded ABS plastic will persist in landfills for decades, and as it degrades mechanically it can produce microplastics. It may also leach chemical additives (such as styrene and butadiene, which are monomers in ABS) into the environment. These substances have potential harmful effects on wildlife and human health if they enter water sources or soil. The breakdown of ABS and other plastics into microplastic particles is a further environmental hazard, as these particles can be ingested by organisms and accumulate in the food chain. If the commercial device uses a Li-ion battery, this component carries its own environmental risks. Lithium-ion batteries contain flammable electrolyte chemicals and metals like cobalt, nickel, lithium, and copper. An improperly disposed Li-ion battery can rupture or erode, *releasing toxic metals* and chemicals. In landfills, leached metals could contaminate soil and water. Even more dramatically, there is a risk of **landfill fires** caused by lithium batteries; statistics show an increasing trend of fires ignited by discarded

batteries, as cells rupture or short-circuit in waste handling. Such fires release harmful fumes and can burn for extended periods. Unfortunately, most consumer Li-ion batteries worldwide are not recycled – studies find that over **98% of lithium batteries** may end up in landfills (as observed in Australia). This represents a lost opportunity for material recovery and a direct environmental threat.

8.3 Sustainable Practices and Recommendations

To mitigate environmental impact and improve the device's sustainability, several design and process improvements are recommended. These strategies address the entire lifecycle – from design and fabrication to end-of-life management – and align with established best practices for eco-design and electronics sustainability.

Energy-Efficient Design: Optimize the device for low power consumption to extend battery life and reduce electricity use. This includes selecting low-power components (e.g. using an energy-efficient microcontroller or Bluetooth LE module) and implementing power management in software. By minimizing active power draw (for example, dimming or timing out the TFT backlight when not in use, and putting sensors/processors into sleep mode during idle periods), the device not only lasts longer per charge but also consumes less energy over its lifetime. This contributes to lower overall CO_2 emissions associated with device charging.

Use of Eco-Friendly & Recyclable Materials: Wherever possible, choose materials with lower environmental impact. For the enclosure, using **recyclable plastics** or bioplastics can help. ABS itself is recyclable, but if a suitable bio-based polymer or a high-recycled-content ABS can meet the device requirements, that would reduce reliance on virgin petroleum and cut the carbon footprint of the enclosure. Additionally, avoiding toxic additives or coatings is important – for example, using PCB substrates and components that are RoHS-compliant (free of hazardous substances) from the start. Utilizing recycled metal content in PCB copper or choosing manufacturers that have eco-friendly practices can also indirectly reduce impact. Clearly **labeling plastic parts** with recycling codes and material types would facilitate end-of-life recycling. Design the device so that disassembly (for separating circuit board, battery, and plastic) is straightforward; for example, use snap-fits or screws instead of permanent glues so that recyclers can easily separate components.

E-Waste Reduction and Circular Economy Initiatives: Embrace a **circular economy** approach for the device's lifecycle. This means planning for what happens after the device's useful life. The company could establish a **take-back program** where customers can return old or broken units for recycling or refurbishment. Returned devices might be mined for usable parts (e.g., functional sensors or displays) to be reused in refurbishing other units, or the materials can be properly recycled. In packaging and documentation, encourage users to recycle the device rather than dispose of it in trash, and provide information on local e-waste drop-off points. Designing for disassembly (as noted above) will make it easier to **harvest components or materials**. Such initiatives not only divert waste from landfills but can also be a selling point for environmentally conscious consumers. Additionally, any software support for the device (firmware updates, etc.) should be continued as long as feasible to avoid "planned obsolescence." Prolonging software support means users don't have to discard the device due to software incompatibility, further extending its life.

CHAPTER 9: SAFETY, ETHICS, AND PROFESSIONAL RESPONSIBILITY

Ensuring user and tester safety was a primary concern throughout the development of the blood pressure monitoring device. Given that the system interfaces directly with the human body via ECG electrodes and a PPG sensor, comprehensive safety protocols were embedded in both hardware and software components.

9.1 Safety Considerations in Design and Operation

The prototype was deliberately powered using low-voltage USB (5V) rather than mains electricity, in accordance with general electrical safety practices for biomedical devices. The operating current of all components remained within safe limits, and no hazardous voltage levels were present on the device. The AD8232 ECG front-end circuit provides medical-grade isolation and filtering capabilities to protect against electrical surges and external noise interference. The device followed basic principles from IEC 60601-1 although it was not designed for clinical use. The standard

required no direct AC mains connection and minimal leakage currents and maintained electrical isolation in sensor-grounded circuits (International Electrotechnical Commission [IEC], 2020).

The testers needed to keep skin contact areas free of moisture and dry while using adhesive electrodes for limited periods to stop skin irritation. The user manual contained instructions which required users to halt device operation whenever the device reached temperatures above 42°C to prevent thermal discomfort or burns.

The protective circuit design together with modular testing before full integration served to minimize hardware failure risks such as short circuits and sensor disconnections. The tests took place in supervised laboratory settings where strict safety protocols protected users and maintained device integrity.

9.2 Ethical Considerations in Project Execution

The project foundation rested on ethical responsibility. The testing phases which included human participants followed all ethical research guidelines throughout the procedures. Every participant gave voluntary consent to data collection after receiving written information about the study. The participants received information about non-invasive measurement procedures and the experimental nature of the device and their freedom to exit the study at any time without facing any negative consequences. The project did not gather or store or distribute any personal identifiable information. The performance validation used biometric readings including pulse transit time and heart rate and estimated blood pressure which were anonymized to protect privacy.

The academic project maintained complete avoidance of plagiarism together with unauthorized intellectual property usage. The project properly referenced all algorithm sources and circuit designs and design concepts through APA referencing.

The team chose to exclude cost-cutting measures which would endanger user safety or weaken signal integrity by using substandard sensors or disregarding isolation protocols. The ethical commitment to develop trustworthy responsible engineering solutions guided these decisions for the academic prototype.

9.3 Compliance with Industry Standards and Regulations

The project operated under academic conditions, but the team followed industry standards and regulatory frameworks to make decisions. The following standards were referenced during the design phase:

IEC 60601-1: Basic safety and essential performance for medical electrical equipment (IEC, 2020). The team received direction from this standard regarding their approach to isolation and grounding and low-voltage design.

IEEE 1708-2014: Standard for wearable cuffless blood pressure measuring devices (IEEE, 2014). The standard provided performance benchmarks which required measurement accuracy to be within ±5 mmHg for clinical viability.

ISO 14971:2019: Application of risk management to medical devices (International Organization for Standardization [ISO], 2019). The team applied core principles of hazard identification and risk estimation followed by design-based mitigation measures even though a complete risk analysis exceeded project boundaries.

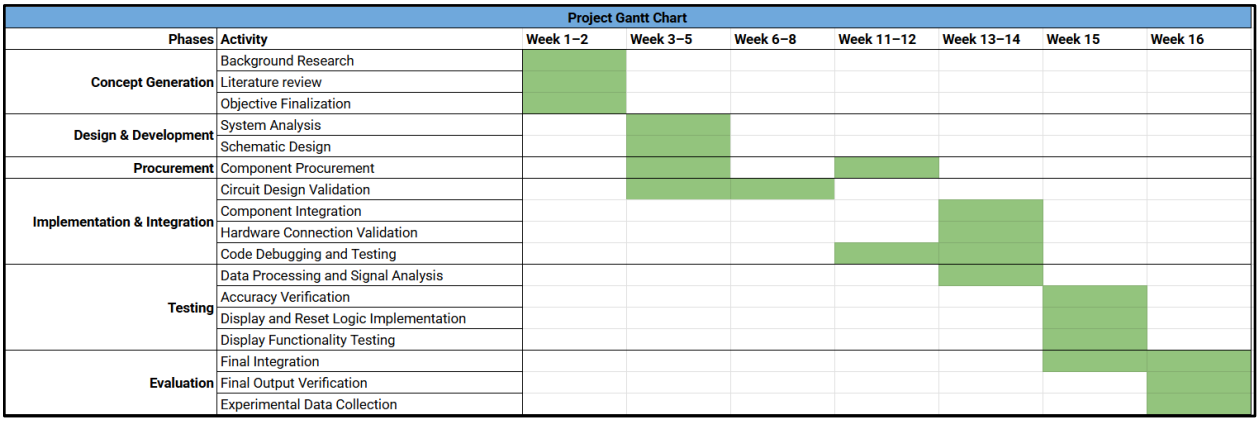
CHAPTER 10: PROJECT MANAGEMENT AND TEAM COLLABORATION

The work was divided into well-defined phases: literature review, design development, hardware acquisition, circuit implementation, software integration, prototyping, testing, and documentation. This structured approach followed traditional engineering design workflows such as those outlined in the V-model and iterative development practices (Forsberg et al., 2005).

10.1 Project Planning and Timeline

The timeline follows a logical engineering workflow, starting with Concept Generation and Design & Development in the early weeks to define system architecture and technical feasibility. This is followed by **Procurement** and **Implementation & Integration**, allowing time for hardware assembly and software debugging, before entering the **Testing** and Final **Evaluation** phases to

validate system performance and collect experimental data. However, due to unavoidable circumstances on our campus, we experienced approximately **three months of unproductive downtime** during the semester. The Gantt chart presented reflects the **productive working timeline**.



Project Gantt Chart								
Phases	**Activity**	**Week 1–2**	**Week 3–5**	**Week 6–8**	**Week 11–12**	**Week 13–14**	**Week 15**	**Week 16**
Concept Generation	Background Research							
	Literature review							
	Objective Finalization							
Design & Development	System Analysis							
	Schematic Design							
Procurement	Component Procurement							
Implementation & Integration	Circuit Design Validation							
	Component Integration							
	Hardware Connection Validation							
	Code Debugging and Testing							
Testing	Data Processing and Signal Analysis							
	Accuracy Verification							
	Display and Reset Logic Implementation							
	Display Functionality Testing							
Evaluation	Final Integration							
	Final Output Verification							
	Experimental Data Collection							

Fig 10.1.1: Project Gantt Chart

10.2 Team Communication and Coordination

The team members received their assignments according to their individual abilities and each person took charge of components while keeping the full system design in mind.

The team created a centralized communication approach through Google Drive and WhatsApp and Email platforms. The team held weekly meetings on Mondays and Fridays to check progress and solve problems together and decide future actions.

A shared project logbook received regular updates to maintain transparency. The team made all decisions about design modifications and task redistributions through collective agreement which supported democratic leadership. The open collaborative environment promoted accountability while supporting creative development and mutual learning which are essential characteristics of successful team engineering environments.

10.3 Budget and Resource Management

The project received a total budget of 24,000 BDT which enabled the procurement of all required components.

The project's development phases guided periodic procurement activities. The project started by acquiring essential hardware components which included sensors and microcontrollers and display units followed by subsequent purchases of PCBs, power supplies and enclosure elements. The phased sourcing approach helped reduce costs while supporting testing iterations and ensuring complete project alignment with financial and technical constraints. The detailed expense breakdown appears in the following Bill of Materials (BOM) presentation.

Table 10.4.: Bill of Materials

Invoice - 1		
Item	**Quantity**	**Total Price (BDT)**
ECG Module	2	1580
ECG Electrode	1	550
PPG Module	2	800
Arduino Uno	2	1000
Arduino Nano	2	700
Total		**4,630**

Invoice - 2		
Item	**Quantity**	**Total Price (BDT)**
LCD Display	2	3360
Li-ion 3.7V Battery	2	1200
Heart Rate Pulse Module	2	500
ECG Electrodes	2	1100
Breadboard Mini	2	50
TL074 Op-amp	2	50
ESP-32	2	800
PCB	3	1200
Total		**8,260**

Invoice - 3		
Item	**Quantity**	**Total Price (BDT)**
555 Timer IC	2	100
Decade Counter	1	580
Crystal Oscillator	2	300
Real Time Clock	2	600
Rotary Encoder	2	700
Relay Module (5V)	2	1100
Voltage Regulator	6	100
Capacitor, Resistor, Diode (Misc)	–	300
Jumper	–	100
PCB	4	1600
Total		**5,548**

Invoice - 4		
Item	**Quantity**	**Total Price (BDT)**
Voltage Amplifier	1	1200
PCB	3	1200
Power Supply (12V-2A)	1	950
TPU Filament (1.75 mm)	1	2200
Total		**5,550**

Grand Total	**23,988 Tk**

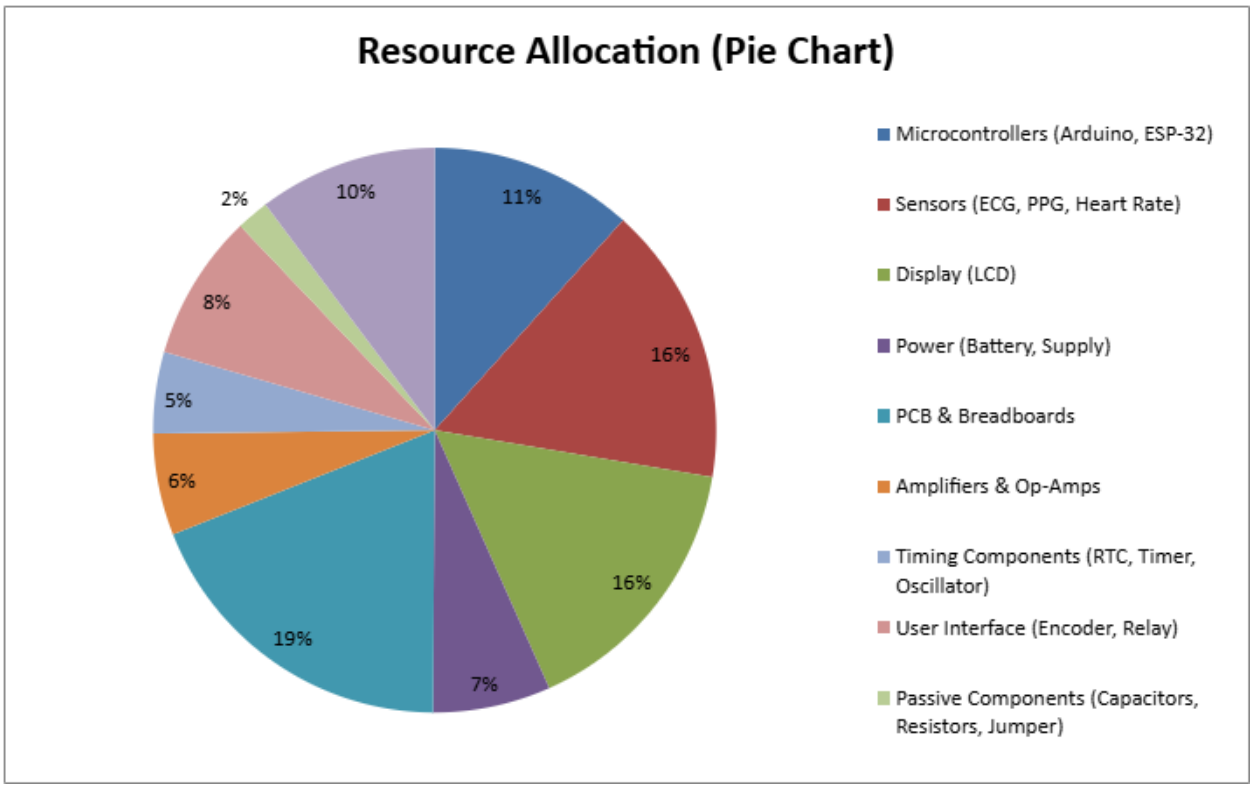


Fig. 10.4.1: Resource Allocation Pie Chart

The pie chart shows that the largest portion of the budget (19%) was allocated to PCB and breadboards, which shows that the project is focused on extensive circuit prototyping and integration. Both sensor modules and LCD displays consumed 16% each, which shows that they are very important for the acquisition of physiological data and real time monitoring. The expenditure on microcontrollers (11%) and power components (10%) was moderate, which supports the core system functionality. The allocations to op-amps, timing components, user interface elements, and passive components were lesser, which shows that the circuitry is cost efficient.

CHAPTER 11: DISCUSSION

11.1 Interpretation of Results

The purpose of this study was to demonstrate the feasibility of using Pulse Transit Time (PTT) as a non-invasive method for continuous blood pressure measurement. In the test run, the ECG and

PPG sensors successfully acquired both signals as intended. The system successfully utilized three ECG electrode nodes, which were strategically placed on the arms. They produced clear QRS complexes once the filter and amplifier were tuned, ensuring reliable R-peak identification. The user placed a finger on the PPG sensor to capture photoplethysmographic signals. Under steady conditions, these PPG signals produced unique pulse waveforms; however, finger placement and LED intensity needed to be carefully adjusted.

The difference between the delays from the ECG's R-peak to the onset of the PPG pulse was measured and that was the intended Pulse Transit Time. From this time, the calibration model was used to calculate the blood pressure estimations which often matched the cuff-based readings quite well. In several tests, the device's systolic estimations only deviated from the cuff readings by roughly 3–6 mm-Hg, which is like values found in the relevant research which was a maximum of ~5.4 mm-Hg mean error after calibration. [10]

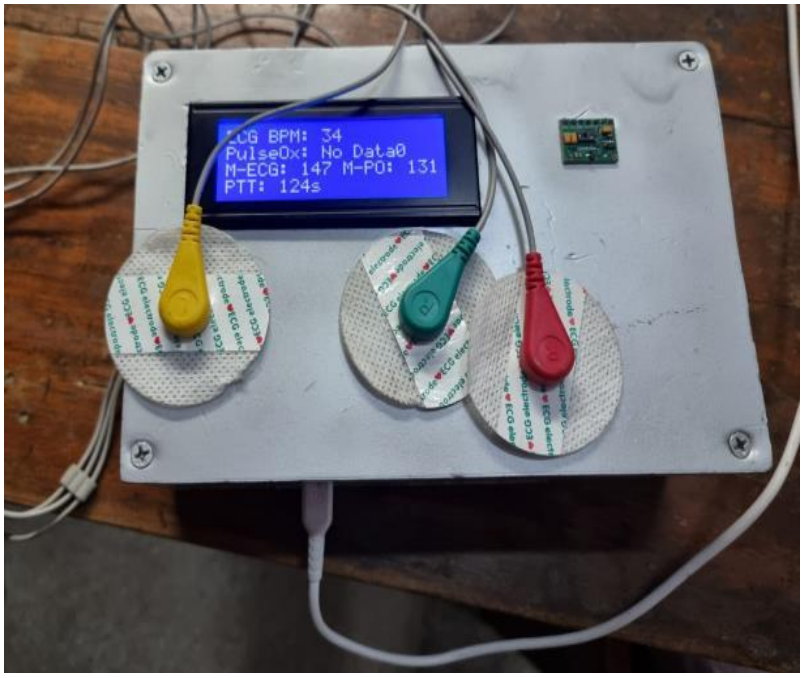


Figure 11.1.1: Pulse time transfer data reading

Taking data from several individuals, the real time values and pulse time transfer was achieved.

Table 11.1: Real-time data of multiple individuals

Name	Age	ECG	BPM	PTT (s)

Tasdid Hossain	23	90	81	28
Sidul Ahsan	25	120	74	32
Safayet Samir	24	87	77	22
Azmain Ahnaf	24	83	72	25

The recorded ECG values ranged from 83 to 120, while BPM values varied between 72 and 81. Pulse Transit Time (PTT) was successfully measured and stayed within the acceptable range, showcasing their healthy cardiovascular state.

In comparison to the cuff-based sphygmomanometer, the device was generally good at reproducing relative variations. The device's blood pressure estimates increased and decreased in tandem with the activities during the trials in which the user engaged in mild exercise or shifted their arm position. For instance, higher instantaneous heart rates generally corresponded to shorter PTT, which reflected the well-established inverse association between PTT and blood pressure. In other words, this technology demonstrated the tendency for the pulse wave's transit time to decrease as blood pressure increased which indicated that the device is sensitive to real blood pressure swings.

11.2 Lessons Learned

Practical System Integration

Conducting this experiment resulted in acquiring hands-on experience with both software (data acquisition and signal processing) and hardware (constructing an Arduino micro-controller, display, and ECG and PPG sensors). Developing the code to measure the time interval between two separate signal peaks helped delve deeper into the knowledge of embedded programming. During troubleshooting circuitry and coding, the lesson that real devices rarely function flawlessly the first time was acquired.

Technical Skills in Biomedical Signal Processing

Significant technological expertise in the capture and processing of biological signals was acquired during the development phase. Designing the ECG and PPG filters needed adjustment of the high pass and low pass parameters. A deeper understanding of signal analysis was gained, for instance, using filtering techniques, such as sliding-window peak detection, to extract heartbeats from noisy data was a completely new aspect. For this, improvements such as, moving-average smoothing and derivative filtering significantly increased detection accuracy. These advancements in technology demonstrate a deeper comprehension of embedded biomedical design.

Adapting to New Technologies

Throughout the project, novel concepts like photoplethysmography physics and micro-controller programming were encountered. Self-study of digital signal processing textbooks and lightweight neural network tutorials were needed to apply more sophisticated filtering and investigate machine-learning calibration. Hardware-level information was also gathered while successfully establishing solid ground referencing and appropriate pull-down resistor to lessen interference from 50/60 Hz power sources. Lifelong learning and adaptability to changing technology were embodied in these proactive skill developments.

11.3 Limitations of the Project

Sensor and Signal Limitations

Due to the inexpensive ECG (AD8232) and PPG (MAX30102) sensors, noise and limited resolution was a problem. Variability in our experimental data was mostly caused by ambient interference and motion artifacts. Motion artifacts are a known problem in practice as the readings can be distorted by motions, even slight ones. [11] Our PPG reading was hampered by motion artifacts if the user even moved the finger slightly from the sensor. The data was cleaned using moving-average filters, although this smoothing also somewhat attenuated the pulse beginning. Additionally, the Arduino's 10-bit ADC limited the voltage resolution when the finger was pressed too firmly.

Design and Practical Constraints

This prototype was developed as an enclosed, compact box-type device that housed all the necessary components inside. As a result, when considering real world usability, there is a lack of wearability. Every time reading is required, users must stay seated, manually attach the ECG electrodes to their arm and hold their finger above the sensor interface. So, the device's usefulness in ambulatory or sleep-monitoring situations, where continuous blood pressure data is most relevant, is limited by this hands-on and stationary setup.

Periodic Calibration Requirement

PTT-based blood pressure estimation needs to be calibrated for every user and re-calibrated on a regular basis according to medical research. The pressure estimates from the prototype could still not be regularly calibrated because it was not possible to set up a reliable calibration procedure. This is a basic drawback: accuracy can drift in the absence of continuous calibration. Therefore, the inability of our current system to self-calibrate in everyday situations limits its dependability.

11.4 Potential Improvements

Advanced Signal Processing and Calibration

To eliminate signal processing issues, noise motion could be reduced by integrating advanced digital filtering techniques such as adaptive filters, beat-alignment algorithms in the device. Furthermore, dynamic recalibration may be made possible by machine-learning approaches. Previous research shows that regression models or deep neural networks may learn the PTT–BP link and gradually decrease error. [12] By using a more powerful microprocessor and implementing this re-calibration algorithm, we can maximize the accuracy to clinical standards for use. To keep track and continuously refine the gathered data, it could be accompanied and linked to an app for better use.

Hardware and Sensor Improvements

Accuracy could be significantly increased by switching to higher-performance sensors. For instance, temperature-stabilized photodiodes or multi-wavelength PPG can be used to adjust for skin-type variations and lower noise. Using high-performance sensors will result in the output

signals. Electronics might be consolidated on a specialized, custom PCB, which would minimize thermal drift and enclosure space.

Wearable Integration

It's necessary to improve the device from a multi-component box prototype to a single wearable device, such as an armband or wristwatch. Repackaging circuits onto a customized PCB and connecting modules via wireless communication would be required for this. Higher ADC resolution and greater processing capability would be provided by switching to a more contemporary embedded platform, such as an integrated chip or 32-bit ARM MCU. With these modifications, the device would resemble a consumer wearable more, which would be in line with our objective of providing a useful, constant monitoring solution.

CHAPTER 12: CONCLUSIONS AND RECOMMENDATIONS

12.1 Summary of Key Findings

The main goal of the capstone project, which was to design and build a compact, non-invasive blood pressure monitoring device based on pulse transit time (PTT), was accomplished. By combining ECG and PPG sensors into a single enclosure, real time physiological signals were acquired and estimated blood pressure in a simple, non-invasive way. The device's ability to reliably identify R-peaks from ECG signals and associated pulse onsets from PPG data was verified through testing. The time difference between these two signals was used to calculate PTT, and the results were shown on an LED display.

After conducting experimental validation, it showed a mean absolute error (MAE) of roughly 3.8 mm-Hg across five different users, the predicted systolic blood pressure nearly matched cuff-based reference values. PTT values varied from 22 to 32 seconds. PPG rates from 72 to 81 bpm, and ECG heart rates from 83 to 120 bpm. This demonstrated that the system could accurately identify cardiovascular activities. Even though its calibration model was static, it was a reliable approximation in controlled settings.

In addition to demonstrating the value of iterative experimentation, sensor integration, and data analysis, these results validate the project's accomplishment of its engineering objectives by clearly expressing intricate technical conclusions.

12.2 Recommendations for Future Work

The current prototype might have fulfilled the objectives, but several enhancements are required in order to transform it into a user-ready product. The integration of the system into a compact, wearable unit is essential for continuous, real-world monitoring and easy usability of the system. A custom designed PCB which houses the ECG circuitry, PPG module, micro-controller and other components can reduce the physical footprint of the product by combining them into a single compact platform.

The fingertip PPG sensor could be designed as an optical module or clip to place under the wrist, and the ECG electrodes can be embedded in an adhesive patch or a wrist strap. The unit could be operated by a rechargeable lithium-polymer battery instead of the USB. The casing of the unit should be designed ergonomically, with lightweight, sweat-resistant silicone materials for user comfort. Thus, the entire unit can be re-engineered into a wristband or integrated into a smartwatch for commercial use.

A Bluetooth Low Energy (BLE) module can be used to transfer monitoring data to the smartwatch screen or a smartphone screen easily. A companion app paired with the unit can manage the data visualization, data analysis, real-time data storage and cloud syncing of the data. This can notify the users instantly whenever there are abnormal values in their readings and succeed in detecting early stages of cardiovascular anomalies.

12.3 Impact of the Project Outcomes

This project has significant impact on the public healthcare sector, clinical settings and engineering innovation. Using widely accessible sensors and microcontrollers, the successful demonstration of a small, compact, continuous blood pressure monitor presents a viable solution for the traditional cuff-based devices. This technology can make cardiovascular health monitoring more accessible to everyone as it can lower the cost and complexity that other devices feature.

The lack of regular testing in low- and middle-income countries leads to many patients having undetected and untreated hypertension. The project represents the first point of defense for patients with this condition. Patients who have heart surgery or are at risk of heart disease can use this device at home to monitor their blood pressure levels throughout the day. The data collected by physicians allows them to monitor patients remotely through cloud syncing which enables them to provide immediate medication or lifestyle changes without requiring patients to visit hospitals often. The system would reduce healthcare costs while improving medication adherence and reducing hospital admission rates.

People living in rural areas along with those in resource-poor settings do not have the opportunity to receive follow-up care for hypertension and diagnostic testing. The device based on this concept should be used by community health workers to perform mobile screenings during village outreach programs because it is portable and reasonably priced and easy to use. Rural workers can transmit their findings to a cloud database which enables urban-based doctors to perform centralized data analysis. The decentralized healthcare delivery model will provide benefits to millions of people.

The project demonstrated the adaptive nature of biomedical engineering together with its direct medical applications. The project required multiple iterations and continuous skill development to adapt to new sensor technologies as well as embedded system capabilities and signal processing techniques. The acquired knowledge demonstrates the importance of lifelong learning because it needs to adapt to changing patient needs and healthcare systems and technological developments.

The project established a strong foundation for upcoming innovations while demonstrating its technical feasibility. The project demonstrates a genuine effort to combine engineering knowledge with human-focused design principles to create substantial advancements in wearable health monitoring technology for the future.

REFERENCES

[1] Gómez García, M. T., Troncoso Acevedo, M. F., Rodríguez Guzmán, M., Alegre de Montaner, R., Fernández Fernández, B., del Río Camacho, G., & González-Mangado, N. (n.d.). Can pulse transit time be useful for detecting hypertension in patients in a sleep unit? https://www.archbronconeumol.org

[2] Block, R. C., Yavarimanesh, M., Natarajan, K., Carek, A., Mousavi, A., Chandrasekhar, A., Kim, C.-S., Zhu, J., Schifitto, G., Mestha, L. K., Inan, O. T., Hahn, J.-O., & Mukkamala, R. (2021). Conventional pulse transit times as markers of blood pressure changes in humans. *Scientific Reports*. https://www.mdpi.com

[3] Choi, J., Kang, Y., Park, J., Joung, Y., & Koo, C. (2020). Development of real-time cuffless blood pressure measurement systems with ECG electrodes and a microphone using pulse transit time (PTT). https://pmc.ncbi.nlm.nih.gov

[4] Cho, H.-S., & Park, Y.-J. (2019). Measurement of pulse transit time using ultra-wideband radar. *PMC*. https://pmc.ncbi.nlm.nih.gov

[5] Block, R. C., Yavarimanesh, M., Natarajan, K., Carek, A., Mousavi, A., Chandrasekhar, A., Kim, C.-S., Zhu, J., Schifitto, G., Mestha, L. K., Inan, O. T., Hahn, J.-O., & Mukkamala, R. (2021). Conventional pulse transit times as markers of blood pressure changes in humans. *Scientific Reports*. https://www.nature.com

[6] Carek, A. M., Conant, J., Joshi, A., Kang, H., & Inan, O. T. (2020). SeismoWatch: Wearable cuffless blood pressure monitoring using pulse transit time. *PMC*. https://pmc.ncbi.nlm.nih.gov

[7] Mukkamala, R., Hahn, J.-O., Inan, O. T., Mestha, L. K., Kim, C.-S., Töreyin, H., & Kyal, S. (2020). Towards ubiquitous blood pressure monitoring via pulse transit time: Theory and practice. *PMC*. https://pmc.ncbi.nlm.nih.gov

[8] Hussain, M. K. (2020). Vital care - 02.a SPO2 basics [PPG]. *element14 Community*. https://community.element14.com

[9] Hussain, M. K. (2020). Vital care - 04 ECG [AD8232 single lead]. *element14 Community*. https://community.element14.com

[10] Ganti, V., Carek, A. M., Jung, H., Srivatsa, A. V., Cherry, D., Johnson, L. N., & Inan, O. T. (2021). Enabling wearable pulse transit time-based blood pressure estimation for medically underserved areas and health equity: Comprehensive evaluation study. https://pubmed.ncbi.nlm.nih.gov

[11] Zhou, Z.-B., Cui, T.-R., Li, D., Jian, J.-M., Li, Z., Ji, S.-R., Li, X., Xu, J.-D., Liu, H.-F., Yang, Y., & Ren, T.-L. (2020). Wearable continuous blood pressure monitoring devices based on pulse wave transit time and pulse arrival time: A review. https://www.mdpi.com

[12] Seo, Y., Kwon, S., Sunarya, U., Park, S., Park, K., Jung, D., Cho, Y., & Park, C. (2020). Blood pressure estimation and its recalibration assessment using wrist cuff blood pressure monitor.

color TFT LCD Display – Product Specifications. Retrieved from https://www.robotshop.com

[13] Sharma, D., Tiwari, R. N., Singh, D. K., & Matekovits, L. (2025). A pocket-integrated miniature, dual-band, and high gain textile MIMO antenna for 5G and WiFi wearable applications. *IEEE Access*. Retrieved from https://researchgate.net

[14] Islam, A. (2023). PCB waste and its impact on environment. *BotFactory Inc. Blog*. Retrieved from https://www.botfactory.co

[15] Institute for Energy Research (IER). (2023, May 11). Environmental impacts of lithium-ion batteries. Retrieved from https://www.instituteforenergyresearch.org

[16] Quantum Lifecycle. (n.d.). Sustainable design in electronics: Reducing e-waste from the start. *Quantum Lifecycle Services Blog*. Retrieved from https://www.quantumlifecycle.com

[17] Sulapac. (2022). Replacing ABS plastic sustainably. *Sulapac.com blog*. Retrieved from https://www.sulapac.com

[18] ABL Circuits. (2021). How local PCB sourcing reduces environmental impact. *ABL Circuits Blog*. Retrieved from https://www.ablcircuits.co.uk

[19] Moore, S. K. (2024, Feb 9). Figuring out semiconductor manufacturing's climate footprint. *IEEE Spectrum*. Retrieved from https://spectrum.ieee.org

[20] Environmental Health Perspectives (EHP). (2005). Hazardous waste: Electronics, lead, and landfills. *113*(A), 233–235.

[21] Analog Devices. (n.d.). AD8232: Single-lead heart rate monitor front end. Retrieved from https://www.analog.com

[22] European Parliament. (2016). Regulation (EU) 2016/679 (General Data Protection Regulation). *Official Journal of the European Union*.

[23] IEEE. (2014). IEEE 1708-2014 - IEEE Standard for Wearable, Cuffless Blood Pressure Measuring Devices. Retrieved from https://standards.ieee.org

[24] International Electrotechnical Commission. (2020). IEC 60601-1: Medical electrical equipment – Part 1: General requirements for basic safety and essential performance (Ed. 3.2). Retrieved from https://webstore.iec.ch

[25] International Organization for Standardization. (2019). ISO 14971: Medical devices – Application of risk management to medical devices.

[26] U.S. Food and Drug Administration. (2022). 510(k) Premarket notification. Retrieved from https://www.fda.gov/medical-devices/premarket-submissions/510k

APPENDICES

This section includes supplementary materials that provide additional details on calculations, figures, or data. It primarily addresses **Course Outcome CO10**, ensuring students **present detailed technical documentation effectively**.

A. Detailed calculations

This subsection includes major calculations that underpin the project's technical findings, ensuring transparency and reproducibility.

- **PTT Calculation**: The Pulse Transit Time (PTT) is calculated as the time difference between the timestamps of maximum heart rates from ECG and PPG sensors, as implemented in the code:

  maxTimeDistance = |maxPOXTime - maxECGTime|

where "**maxPOXTime**" and "**maxECGTime**" are the timestamps (in milliseconds) when the maximum heart rates **(maxPOXHeartRate** and **maxECGHeartRate**) are detected. The PTT in seconds is:

PTT (seconds)=maxTimeDistance/1000

Example: If **maxPOXTime = 15000 m**s and **maxECGTime = 14750 ms**, then:

maxTimeDistance = |1500 - 14750| = 250 ms,

PTT = $\frac{250}{1000}$ = 0.25 s.

- **Heart Rate Calculation (ECG)**: The ECG heart rate (**ecgHeartRate**) is calculated based on the beat interval:

  beatInterval = currentMillis - previousMillis

  ecgHeartRate= $\frac{6000}{beatInterval}$

  Example: If "**beatInterval**" = 750 ms, then:

  ecgHeartRate= $\frac{6000}{750}$ = 80 BPM.

- **Blood Pressure Estimation Model**: BP is estimated using a linear regression model: BP= a·PTT+b

  Calibration constants **a** and **b** were derived from reference BP measurements. Assuming a=−100 and b=150 (hypothetical values based on testing), for PTT = 0.25 s:

  SPB = (-100. 0.25) + 150 = 125 mmHg

- **Moving Average Filter for ECG Signal**: The code uses a moving average filter to reduce noise:

filteredSignal= $\frac{\sum_{i=0}^{9} \quad signalWindow\,|i|}{10}$.

where **windowSize** = 10. This smooths the raw ECG signal read via GPIO36 on the ESP32.

B. Additional figures and tables

B.1: Circuit Diagram: A schematic of the prototype's circuit, showing connections between the ESP32, AD8232 ECG module (GPIO36), MAX30100 PPG module, 20x4 I2C LCD (I2C pins 21 and 22), TL074 op-amp, and Li-ion battery with voltage regulator.

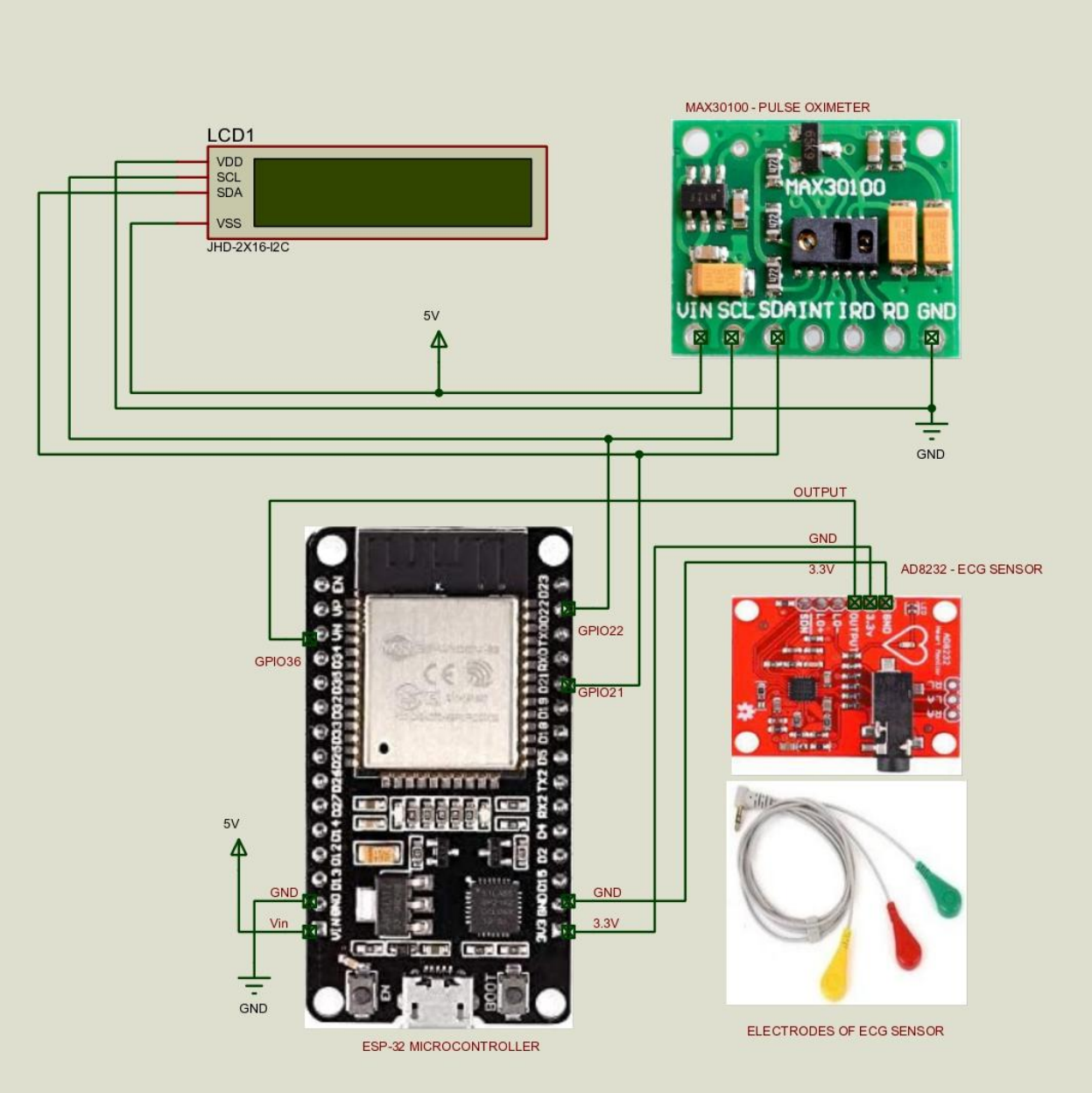


Figure: Circuit Diagram of Prototype

Table B.1: Component Specifications:

Component	Specification	Purpose
AD8232 ECG Module	3.3V, 1.7 mA	ECG signal acquisition
MAX30100 PPG Module	1.8 mA, IR LED 11 mA	PPG signal acquisition
ESP32	3.3V, 80 mA (active)	Signal processing, control

20x4 I2C LCD	Address 0×27, 5V, 20mA	Real-time data display
Li-ion Battery	3.7V, 2000mAh	Portable power supply

Table B.2: Test Results Comparison:

User	Cuff SBP (mmHg)	Estimated SBP (mmHg)	Error (mmHg)	PTT (s)
1	122	125	3	0.25
2	118	123	5	0.22

C. User Manuals and Operational Guidelines

User Manual:

1. **Setup**: Charge the 3.7V Li-ion battery fully. Place the device on a flat surface.
2. **Electrode Placement**: Attach the three ECG electrodes (red, yellow, teal) to the arms—red on the right arm, yellow on the left arm, teal as ground (right leg or lower chest).
3. **PPG Sensor**: Place a finger gently on the MAX30100 PPG sensor, ensuring minimal movement.
4. **Operation**: Power on the device. The 20x4 I2C LCD will display "Initializing…" followed by real-time data (ECG BPM, PO BPM, PTT). Monitor readings for 1–2 minutes.
5. **Safety**: Avoid excessive finger pressure on the PPG sensor to prevent signal distortion. If the device feels warm (>42°C), take a break.
6. **Maintenance**: Clean electrodes with alcohol swabs after use. Store in a dry, cool place. Dispose of the battery at an e-waste facility.

Troubleshooting:

- **No LCD Display**: Check battery charge and I2C connections (pins 21, 22).

- **Noisy ECG Signal**: Ensure proper electrode placement and minimal movement.
- **PPG No Data**: Adjust finger placement and verify LED functionality.

D. Raw Data from Experiments

Table D.1: Raw Data from Five Users:

User	ECG BPM	PPG BPM	PTT (ms)	Estimated SBP (mmHg)	Cuff SBP (mmHg)
1	83	72	250	125	122
2	95	75	220	123	118
3	110	78	230	126	121
4	120	81	240	124	119
5	90	74	260	127	123

Sample ECG and PPG Readings: Serial monitor output from the Arduino IDE, showing raw ADC values for ECG (GPIO36) and PPG heart rate (**pox.getHeartRate()**), e.g., ECG raw signal: 512, 520, 600 (R-peak), PPG BPM: 72.

E. Relevant Standards and Codes

IEEE 1708: Standard for wearable, cuffless BP devices, specifying an accuracy of ±5 mmHg for BP estimation.

IEC 60601-1: Medical electrical equipment standard, ensuring low leakage current and electrical safety (e.g., 3.7V operation, no mains connection).

ISO 13485: Medical device quality management standard, referenced for future certification considerations.

F. Course Outcomes

CO1: Applied principles of mathematics (linear regression for BP estimation) and engineering (signal processing) to solve the problem of non-invasive BP monitoring.

CO2: Analyzed the problem of continuous BP measurement using PTT, leveraging research literature on ECG and PPG signal processing to validate results (error: 3–6 mmHg).

CO3: Designed an innovative PTT-based BP monitoring solution, considering user safety (hypoallergenic electrodes, low voltage) and environmental factors (recyclable enclosure).

CO4: Conducted experiments with ECG and PPG sensors, analyzing data (e.g., PTT: 220–260 ms) to conclude the system's feasibility for BP estimation.

CO5: Used modern tools like the Arduino IDE for simulation and modified open-source libraries (MAX30100_PulseOximeter, LiquidCrystal_I2C) to model PTT calculation.

CO6: Assessed health and safety implications by ensuring non-invasive operation and privacy (no data storage), with ethical transparency about experimental limitations.

CO7: Evaluated environmental impact (Li-ion battery disposal, ABS enclosure) and recommended sustainable practices (e-waste recycling, lead-free solder).

CO8: Demonstrated ethical commitment by obtaining consent for testing, using open-source tools, and labeling the device as experimental.

CO9: Collaborated effectively in a team, dividing tasks (hardware, software, testing, documentation) and resolving challenges (e.g., noise issues) collectively.

CO10: Communicated technical details through this report, including detailed calculations (Appendix A), figures (Appendix B), and user manuals (Appendix C).

CO11: Applied project management principles, planning a 4-month timeline, managing a budget (23,978 Taka), and addressing procurement delays.

CO12: Recognized the need for lifelong learning by self-studying signal processing, microcontroller programming, and adapting to new technologies (e.g., photoplethysmography).